Continuing the Alaoglu–Erdős Program

Synchronized Sturmian Dynamics, Effective Logarithmic Separation, and Sharp p-Adic Determinant Ceilings

A continuation of the attached unified report on

https://github.com/VladimirReshetnikov/ProveIt

Verdict. I did not obtain an unconditional proof of the Alaoglu–Erdős conjecture. The exact 2×2 logarithmic determinant remains the terminal obstruction. The continuation does, however, produce several rigorous advances and two fairly sharp no-go theorems for the most natural determinant strategies.

A hypothetical primitive counterexample has a new exact description as one aperiodic Sturmian word simultaneously driving a dyadic and a triadic rational recurrence. Baker's theory upgrades the elementary exponential continued-fraction bound in the source report to an effective polynomial irrationality measure for the primitive generator. On the p-adic side, ordinary block Vandermonde valuations can be computed through all primes in the fixed supports, giving a depth-sensitive asymptotic and a universal $87.1049\ldots\%$ ceiling for the product of the input and output determinants. For the balanced mixed-semigroup determinant, the exact asymptotic min-plus divisor is governed by a two-dimensional Gini norm. The resulting universal ceiling is $4/5$; it improves to $7/10$ in the cross-coprime branch, while the structural places 2 and 3 alone contribute only $3/10$ there. Finally, using the complete triangular (n, k) grid does not remove the termwise obstruction: optimal transport gives the universal ceiling $\pi/4$ relative to the sharp largest-term archimedean bound.

These results identify the remaining opportunity precisely. A determinant proof must obtain leading-order divisibility from genuine cancellation among equal-valuation terms, not merely from entrywise valuations; or it must beat the standard archimedean bound through the Harish–Chandra–Itzykson–Zuber free energy. The report gives exact formalization targets for both possibilities.

Prepared for Vladimir Reshetnikov

OpenAI GPT-5.6 Pro

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Abstract

Let $\vartheta = \log 3/\log 2$. The Alaoglu–Erdős conjecture says that $2^x, 3^x \in \mathbb{Z}$ for real x forces $x \in \mathbb{Z}$. The attached unified report develops an exact conditional structure: if the conjecture fails, there is a least nonintegral generator β, integers $M = 2^\beta$ and $A = 3^\beta$, and unique coordinate descriptions $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ and $\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}$. It also gives local and mixed Vandermonde determinant bounds, but finds a missing logarithm between the number of nodes available in a dyadic block and the number required by a purely archimedean interpolation argument.

This continuation first normalizes $\beta = B + \alpha$, $0 < \alpha < 1$, and proves that failure is equivalent to a single irrational lower mechanical word $s_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$ simultaneously driving

$$R_{k+1} = \frac{(M/2^B)R_k}{2^{s_k}}, \qquad S_{k+1} = \frac{(A/3^B)S_k}{3^{s_k}},$$

with $R_k \in [1, 2)$ and $S_k \in [1, 3)$. This is a synchronized dyadic–triadic Sturmian formulation, and the primitive-output theorem makes at least one denominator sequence attain its full base-power depth. Next, Baker–Wüstholz/Matveev lower bounds give effective constants $c(M) > 0$ and $\mu(M) < \infty$ for which $|\beta - p/q| \geq c(M)q^{-\mu(M)}$; consequently the convergent denominators grow only polynomially from one stage to the next. This is much stronger than the elementary single-exponential recurrence in the source report, although still compatible with the exact conditional block population.

The main new work is a sharp p-adic audit. For the ordinary Vandermonde on all conditional candidates in one dyadic block, all valuations at primes supporting M can be evaluated exactly, and the analogous output formula is exact up to $O(T^2)$. If $a = v_2(M)$, $b = v_3(A)$, $\delta_2 = \beta - a$, and $\delta_3 = \beta - b$, then the all-support termwise divisor of the product of the input and output Vandermondes occupies the asymptotic fraction

$$1 - \frac{\delta_2 \log 2 + \delta_3 \log 3}{3\beta \log 6}$$

of the leading archimedean budget. Since the primitive pair has $\min(a, b) = 0$, this fraction is at most $1 - \log 2/(3 \log 6) = 0.871049\ldots$.

For the mixed-semigroup determinant, let (u_j, v_j) be the first N points of $\mathbb{N}_0^2$ ordered by $u + \vartheta v$. The first-order assignment minima at every prime admit an exact continuum limit. If $X = (X_1, X_2)$ is uniform on the triangle $\{x, z \geq 0 : x + z \leq 1\}$, define the Gini norm $\mathfrak{G}(r, s) = \mathbb{E}|r(X_1 - X_1') + s(X_2 - X_2')|$. Normalized prime-valuation slope vectors $c_p \in \mathbb{R}^2$ satisfy $\sum_p c_p = 0$, and the termwise divisor fraction is

$$1 - \frac{3}{8} \sum_p \mathfrak{G}(c_p) + o(1).$$

The primitive condition and the triangle inequality give $\sum_p \mathfrak{G}(c_p) \geq 8/15$, hence the universal ceiling $4/5$. If $\gcd(MA, 6) = 1$, the ceiling improves to $7/10$; using only $p = 2, 3$ gives $3/10$. The Gini norm is evaluated in closed form from the three vertex values $0, r, s$. This corrects a subsequent repository note whose simultaneous base-depth normalization and proposed all-prime $2/3$ ceiling are not valid in full generality.

Finally, the complete triangular grid of all (n, k) with $n + k\beta \leq T$ supplies $\asymp T^2$ rows. Nevertheless, termwise divisibility still has a universal transport ceiling. The normalized row heights and mixed column weights both have quantile $Q(t) = \sqrt{t}$; therefore the smallest and largest assignment energies tend to $\pi/8$ and $1/2$, respectively. Any all-prime min-plus divisor is bounded by the former, whereas the sharp largest-term determinant bound is governed by the latter. Their ratio is $\pi/4$. The only remaining determinant mechanism is thus genuine same-valuation cancellation or a sharper archimedean free-energy estimate. The Harish–Chandra–Itzykson–Zuber identity is recorded as the natural analytic framework, and the report ends with a branch-sensitive research and Lean formalization program.

Contents

1 Executive answer, scope, and trust boundaries

1.1 The requested goal and the outcome

The requested primary goal was to continue the attached research program and prove the following statement.

Conjecture 1.1 (Alaoglu–Erdős)**.** *If $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

I did not find an unconditional proof. In particular, none of the arguments below converts

$$\log(2^x)\log 3 - \log(3^x)\log 2 = 0$$

into a forbidden linear form in logarithms of algebraic numbers. The equality is bilinear in logarithms, and normalizing it gives exactly the unresolved 2×2 determinant behind the four exponentials conjecture. Baker-type estimates control every *nonzero* linear form that occurs after fixing integer coefficients; they do not prove that this special bilinear form cannot vanish.

The continuation nevertheless changes the technical map in useful ways. It obtains an exact symbolic-dynamical formulation, replaces an elementary continued-fraction estimate by an effective polynomial one, computes the full leading p-adic content of ordinary block Vandermondes, and gives sharp continuum formulas for the termwise divisibility of two much larger determinant constructions. These calculations do more than say that a method "seems too weak": they quantify the leading constant that is available and isolate the precise kind of extra cancellation a successful proof would have to produce.

1.2 The principal new results

Result	Content	Status
Synchronized Sturmian orbit	A counterexample is equivalent to one irrational mechanical word simultaneously controlling the dyadic and triadic denominator jumps of two compact rational recurrences.	**[new paper proof]**
Effective irrationality of β	Baker–Wüstholz/Matveev gives $\lvert\beta - p/q\rvert \geq c(M)q^{-\mu(M)}$ and hence polynomial, rather than exponential, recurrence bounds for continued-fraction denominators.	**[classical]** plus **[new paper proof]**
Ordinary Vandermonde audit	The valuations at every prime in the fixed supports of M and A are computed to leading order. The product construction has universal ceiling $1 - \log 2/(3\log 6) = 0.871049\ldots$.	**[new paper proof]**
Gini assignment formula	For the balanced mixed-semigroup determinant, the exact termwise all-prime divisor fraction is $1 - \frac{3}{8}\sum_p \mathfrak{G}(c_p) + o(1)$ for an explicit norm $\mathfrak{G}$.	**[new paper proof]**

Result	Content	Status
Universal 4/5 ceiling	The primitive condition forces $\sum_p \mathfrak{G}(c_p) \geq 8/15$. Thus entrywise/min-plus divisibility can supply at most 80% of the leading block determinant height.	[new paper proof]
Cross-coprime constants	If $\gcd(MA, 6) = 1$, the all-prime ceiling is $7/10$, and the places 2 and 3 alone contribute exactly $3/10$ in the continuum limit.	[new paper proof]
Full-grid transport ceiling	On the complete triangular (n, k) grid, any termwise all-prime divisor is bounded by the countermonotone energy $\pi/8$, while the largest determinant term has energy $1/2$; the ratio is $\pi/4$.	[new paper proof]
HCIZ reformulation	The full-grid archimedean problem is expressed by the Harish–Chandra–Itzykson–Zuber integral, identifying a concrete free-energy inequality as the remaining analytic target.	[classical] plus [new paper proof]

1.3 Evidence labels

The labels used in this document are intentionally conservative.

[Lean] The attached report says the statement has been accepted by the Lean kernel in the repository snapshot it audits. I did not independently replay the entire Lean build in this environment.

[repository build] A current repository commit or report states that an aggregate build passed. This is repository metadata, not an independently reproduced kernel audit here.

[new paper proof] A complete mathematical argument is supplied in this report. It has been algebraically and numerically stress-tested, but it is not yet formalized or peer-reviewed.

[classical] The result is standard literature, used with an explicit citation.

[computation] A finite calculation or convergence table checks formulas and constants; it is not used as a substitute for proof.

[open] The statement is proposed as a precise target, not asserted.

1.4 What the determinant ceilings do and do not prove

A ceiling such as $4/5$ has a specific meaning. Given an integer determinant $D = \det(E_{ij})$, define at each prime p the minimum valuation of a Leibniz term,

$$\tau_p = \min_{\sigma} \sum_i v_p(E_{i,\sigma(i)}).$$

Every term, and therefore D, is divisible by p^{τ_p}. The product $\prod_p p^{\tau_p}$ is the *termwise* or *tropical* divisor. It ignores extra valuation caused by cancellation among several terms attaining the same minimum.

The 4/5 theorem says that this automatic divisor cannot occupy more than four-fifths of the leading archimedean height for the balanced block construction. It does *not* say that the true valuation $v_p(D)$ is at most the tropical minimum; the opposite inequality holds, and extra cancellation may be large. Thus the result retires only the naïve entrywise-divisibility version of the method. It leaves a sharply defined cancellation problem.

2 Baseline inherited from the unified report

2.1 Notation

Set

$$\vartheta = \frac{\log 3}{\log 2}.$$

The solution and candidate sets are

$$\mathcal{S} = \{x \geq 0 : 2^x, 3^x \in \mathbb{N}\}, \qquad \mathcal{C} = \{m \in \mathbb{N} : m^\vartheta \in \mathbb{N}\}.$$

The map $x \mapsto 2^x$ is a bijection $\mathcal{S} \to \mathcal{C}$. Integer exponents give the obvious solutions $\mathbb{N}_0 \subseteq \mathcal{S}$ and candidates $2^{\mathbb{N}_0} \subseteq \mathcal{C}$.

Throughout the conditional sections, suppose the conjecture fails. The exact structure proved in the source report then supplies a least nonintegral solution $\beta > 0$ and primitive outputs

$$M = 2^\beta \in \mathbb{N}, \qquad A = 3^\beta \in \mathbb{N}.$$

Every solution and candidate has a unique coordinate representation

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}, \qquad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}. \tag{1}$$

The associated output is

$$(2^n M^k)^\vartheta = 3^n A^k.$$

We write

$$a = v_2(M), \qquad b = v_3(A), \qquad c = v_2(A), \qquad d = v_3(M).$$

The inherited primitive-output theorem includes the crucial condition

$$\min(a, b) = 0. \tag{2}$$

It is important not to silently strengthen (2) to $a = b = 0$.

2.2 Inherited results used here

The following facts are taken from the attached unified report and its referenced Lean modules.

- **[Lean]** A nonintegral solution, if it exists, is transcendental. The constant ϑ is transcendental, by Gelfond–Schneider.
- **[Lean]** Failure gives the exact free rank-two monoid structure (1), including uniqueness of (n, k).

- [Lean] In a dyadic exponent block $[T, T+1)$, the solutions are exactly

$$T \quad \text{and} \quad T - \lfloor k\beta \rfloor + k\beta, \qquad 1 \le k \le \left\lfloor \frac{T+1}{\beta} \right\rfloor.$$

  Thus the block count is $1 + \lfloor (T+1)/\beta \rfloor$.
- [Lean] The primitive pair obeys (2); both outputs cannot simultaneously be $\{2,3\}$-smooth.
- [Lean] Mixed monomials $m^u(m^\vartheta)^v$ are integers for $(u,v) \in \mathbb{N}_0^2$, and generalized Vandermonde determinants built from distinct candidate nodes and distinct exponents $u + \vartheta v$ are nonzero integers.
- [Lean] The kernel-checked finite exclusion in the attached report gives $x \ge 12$ for a nonintegral solution. The report separately records a directed-rounding software exclusion through $x \le 26$.

The continuation does not reprove this large inherited corpus. It restates every inherited fact at the point where it enters a new argument.

2.3 Repository reconciliation as of August 21, 2026

The attached source is internally newer than one bibliographic pin printed near its end: its body and status matrix already discuss the enlarged algebraic-output and determinant module surface, although a repository item still names commit `5d3982ceb12644fdd3499569d9db5f62f9a31dc6`. During this continuation I checked the current public repository history. Commit `78f80c5d3f6f9afd64ecb72ddb154c85e2036bc1` reports a successful aggregate build with 67 imported modules and merges the later algebraic-output work. Subsequent commits add a literature compass and a short p-adic priority note; the latest snapshot consulted while writing was `ff9a9f5aa3a3f02599de93fc053e60918f269855`.

Status. [repository build]

The build claims in the preceding paragraph are taken from repository commit messages and files. I did not independently run all Lean jobs. The mathematical arguments newly stated below are therefore labelled [new paper proof], not [Lean].

2.4 The exact remaining logarithmic wall

Writing $M = 2^\beta$ and $A = 3^\beta$ gives

$$\beta = \frac{\log M}{\log 2} = \frac{\log A}{\log 3}$$

and hence

$$\log M \, \log 3 - \log A \, \log 2 = 0. \tag{3}$$

The four numbers $\log 2, \log 3, \log M, \log A$ are logarithms of algebraic numbers. Equation (3) is a vanishing 2×2 determinant. Four exponentials would exclude it because $2, 3$ are multiplicatively independent and $1, \beta$ are linearly independent over $\mathbb{Q}$, but four exponentials remains open. The new work below therefore pursues information that is invisible in the bare real logarithmic determinant: exact floor coding, effective separation, and valuations of large integer determinants.

3 A synchronized Sturmian model of a counterexample

3.1 Fractional-part normalization

Write

$$\beta = B + \alpha, \qquad B = \lfloor \beta \rfloor \in \mathbb{N}_0, \qquad 0 < \alpha < 1.$$

Since a nonintegral solution is transcendental, α is irrational. Define the rational localized multipliers

$$U = \frac{M}{2^B} = 2^\alpha \in \mathbb{Q} \cap (1,2), \qquad V = \frac{A}{3^B} = 3^\alpha \in \mathbb{Q} \cap (1,3). \tag{4}$$

For $k \geq 0$, put

$$e_k = \lfloor k\alpha \rfloor, \qquad s_k = e_{k+1} - e_k \in \{0,1\}.$$

The binary sequence $s = s_0 s_1 s_2 \cdots$ is the lower mechanical word of slope α. Define compact normalized orbits

$$R_k = \frac{U^k}{2^{e_k}} = 2^{\{k\alpha\}} \in [1,2), \qquad S_k = \frac{V^k}{3^{e_k}} = 3^{\{k\alpha\}} \in [1,3). \tag{5}$$

Both orbits use the *same* fractional part and hence the same symbolic itinerary.

Theorem 3.1 (Synchronized dyadic–triadic Sturmian recurrence)**.** *Assume a primitive nonintegral solution β exists. Then the rational sequences R_k, S_k in (5) satisfy*

$$R_0 = S_0 = 1, \qquad R_{k+1} = \frac{UR_k}{2^{s_k}}, \qquad S_{k+1} = \frac{VS_k}{3^{s_k}}, \tag{6}$$

where

$$s_k = 1 \iff \{k\alpha\} \geq 1 - \alpha \iff UR_k \geq 2 \iff VS_k \geq 3. \tag{7}$$

Moreover,

$$\sum_{j=0}^{N-1} s_j = \lfloor N\alpha \rfloor \qquad (N \geq 1). \tag{8}$$

Proof. The recurrence follows by subtracting consecutive floor exponents:

$$R_{k+1} = \frac{U^{k+1}}{2^{e_{k+1}}} = \frac{UU^k}{2^{e_k + s_k}} = \frac{UR_k}{2^{s_k}},$$

and the same calculation with base 3 gives the formula for S_k. Since $e_{k+1} = e_k + 1$ exactly when $\{k\alpha\} + \alpha \geq 1$, the first equivalence in (7) holds. The identities $UR_k = 2^{\alpha + \{k\alpha\}}$ and $VS_k = 3^{\alpha+\{k\alpha\}}$ give the other two. Finally, (8) telescopes: $\sum_{j<N} s_j = e_N - e_0 = \lfloor N\alpha \rfloor$. □

3.2 Sturmian consequences

The following facts are classical consequences of irrational mechanical coding [9, 10].

Corollary 3.2 (Aperiodicity, balance, and minimal complexity)**.** *The word s in Theorem 3.1 is aperiodic. Any two factors of equal length contain numbers of 1's differing by at most one, and the number of distinct factors of length n is exactly $n + 1$.*

Aperiodicity has an elementary proof in this setting. If s were eventually periodic, then its limiting density of 1's would be rational. Equation (8) shows that this density is α, contradicting irrationality. Balance and factor complexity are the standard Sturmian characterization.

The point is not merely terminological. A hypothetical counterexample is no longer described only by two rational numbers U, V satisfying a logarithmic equality. It determines one minimal-complexity aperiodic word, and that same word must be realized by two rational piecewise-multiplicative systems with multiplicatively independent reset bases 2 and 3. This is a much more rigid synchronized object.

3.3 Exact reduced denominators

The primitive factor depths determine how much cancellation occurs in (5). Write

$$M = 2^a W, \quad W \text{ odd}, \qquad A = 3^b Z, \quad 3 \nmid Z.$$

Because M is not a power of 2 and A is not a power of 3, one has $W, Z > 1$.

Proposition 3.3 (Denominator exponents and common jump word)**.** *In lowest terms,*

$$R_k = \frac{W^k}{2^{q_k^{(2)}}}, \qquad q_k^{(2)} = \lfloor k\beta \rfloor - ak = k(B-a) + \lfloor k\alpha \rfloor, \tag{9}$$

$$S_k = \frac{Z^k}{3^{q_k^{(3)}}}, \qquad q_k^{(3)} = \lfloor k\beta \rfloor - bk = k(B-b) + \lfloor k\alpha \rfloor. \tag{10}$$

Consequently

$$q_{k+1}^{(2)} - q_k^{(2)} = B - a + s_k, \qquad q_{k+1}^{(3)} - q_k^{(3)} = B - b + s_k.$$

At least one of the two sequences has baseline jump B, because $\min(a, b) = 0$.

Proof. Using $\lfloor k\beta \rfloor = kB + \lfloor k\alpha \rfloor$,

$$R_k = \frac{M^k}{2^{\lfloor k\beta \rfloor}} = \frac{2^{ak} W^k}{2^{\lfloor k\beta \rfloor}} = \frac{W^k}{2^{\lfloor k\beta \rfloor - ak}}.$$

The numerator is odd, so this is reduced. The base-3 calculation is identical. Taking successive differences gives the jump formulas, and the last claim is the inherited primitive condition (2). ☐

Thus the same Sturmian bit s_k records whether *both* reduced denominators take their large or small permitted jump at time k. One denominator can have a shallower constant baseline, but the aperiodic fluctuation is synchronized exactly.

3.4 An exact equivalent exclusion principle

The source report already contains localization reformulations such as $2^\alpha \in \mathbb{Z}[1/2]$ and $3^\alpha \in \mathbb{Z}[1/3]$. The recurrence packages them into a symbolic statement.

Theorem 3.4 (Synchronized Sturmian exclusion is equivalent to Alaoglu–Erdős)**.** *The Alaoglu–Erdős conjecture is equivalent to the nonexistence of data*

$$\alpha \in (0,1) \setminus \mathbb{Q}, \qquad U \in \mathbb{Q} \cap (1,2), \qquad V \in \mathbb{Q} \cap (1,3)$$

such that, for the lower mechanical word $s_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$*, the recurrences*

$$R_0 = S_0 = 1, \qquad R_{k+1} = UR_k/2^{s_k}, \qquad S_{k+1} = VS_k/3^{s_k}$$

remain in $[1,2) \cap \mathbb{Q}$ *and* $[1,3) \cap \mathbb{Q}$*, respectively, for every* k*, with* $U = 2^\alpha$ *and* $V = 3^\alpha$*. Equivalently, it suffices to rule out irrational* α *for which*

$$2^\alpha \in \mathbb{Z}[1/2], \qquad 3^\alpha \in \mathbb{Z}[1/3].$$

Proof. A counterexample $\beta = B + \alpha$ gives the data by Theorem 3.1. Conversely, if such α, U, V exist, choose an integer B at least as large as the powers of 2 and 3 occurring in the respective reduced denominators of U and V. Then

$$2^{B+\alpha} = 2^B U \in \mathbb{N}, \qquad 3^{B+\alpha} = 3^B V \in \mathbb{N},$$

while $B + \alpha \notin \mathbb{Z}$. The recurrence identities are equivalent to $R_k = 2^{\{k\alpha\}}$ and $S_k = 3^{\{k\alpha\}}$, so they add an exact dynamic coding but no hidden hypothesis. □

3.5 Cobham-type and Mahler-type diagnostics

Cobham's theorem says that a sequence automatic in two multiplicatively independent integer bases is ultimately periodic [11, 12]. Sturmian words of irrational slope are not automatic in any integer base. Theorem 3.4 therefore suggests a tempting strategy: show that the dyadic rational recurrence makes s 2-automatic and that the triadic recurrence makes it 3-automatic.

That implication is *not* presently available. Rationality of the multiplier does not turn the compact rational state R_k or S_k into a finite-state object; the exact reduced denominators grow without bound. Cobham's theorem becomes applicable only after proving a finite-kernel or finite-substitution property, and that is essentially a new arithmetic statement. The synchronized formulation is useful because it identifies this missing lemma cleanly rather than treating "automaticity" as a metaphor.

Research target 3.5 (Finite-kernel extraction)**.** Show that simultaneous rational realizability of the same irrational mechanical word by (6) forces a finite 2-kernel and a finite 3-kernel, or derive a weaker pair of incompatible substitutive structures.

A Mahler-function variant is also conceivable: encode the jump word by $F(z) = \sum_{k \geq 0} s_k z^k$. Aperiodicity and the finite coefficient set already imply, by classical results of Fatou type, that F is transcendental over $\mathbb{Q}(z)$. A proof would have to derive incompatible functional equations from the two rational recurrences. No such equations were found here; the recurrence is multiplicative in the state, not self-similar in the index.

4 Effective logarithmic separation

4.1 From linear forms in two logarithms to an irrationality measure

The source report proves an elementary denominator-growth statement: a very good rational approximation to a hypothetical solution β would force a correspondingly large output, and consecutive continued-fraction denominators obey a single-exponential recurrence. Since β is itself a ratio of logarithms of fixed integers, Baker's theory gives a substantially stronger conclusion.

Theorem 4.1 (Effective finite irrationality measure for the primitive generator)**.** *Assume a primitive counterexample β exists, and put $M = 2^\beta$. There are effectively computable constants $c_M > 0$ and $\mu_M < \infty$ such that, for every $p \in \mathbb{Z}$ and $q \geq 2$,*

$$\left|\beta - \frac{p}{q}\right| \geq c_M q^{-\mu_M}. \tag{11}$$

One may choose μ_M bounded by an effectively computable absolute constant times $1 + \log M$, hence by an absolute constant times $1 + \beta$. The same conclusion follows from $\beta = \log A/\log 3$.

Proof. Consider the linear form

$$\Lambda = q \log M - p \log 2.$$

It is nonzero: otherwise $M^q = 2^p$, which would make $\beta = p/q$ rational; a rational solution is integral by the elementary rational-exponent classification already formalized in the source corpus. The Baker–Wüstholz theorem, or Matveev's explicit lower bound for a homogeneous linear form in logarithms of algebraic numbers, gives

$$\log |\Lambda| \geq -C\,(1 + \log M) \log H - C'(1 + \log M),$$

where $H \geq \max(3, |p|, q)$ and C, C' are effective absolute constants after the fixed number 2 is absorbed. For approximations with $|p| \leq (\beta + 1)q$, one has $H \ll_M q$; for all other p the desired inequality is trivial after decreasing c_M. Since

$$\left|\beta - \frac{p}{q}\right| = \frac{|\Lambda|}{q \log 2},$$

absorbing fixed factors and one additional power of q gives (11). Matveev's dependence on the logarithmic heights is linear in $\log M = \beta \log 2$ here, which gives the final scale assertion. ☐

Status. [classical] plus **[new paper proof]**

The lower-bound theorem for logarithmic forms is classical [2, 3, 4]. Its application to the primitive generator, the continued-fraction consequences below, and the comparison with the source report are new deductions in this continuation. No useful small numerical value of μ_M is claimed; standard explicit constants are very large.

4.2 Polynomial recurrence for convergent denominators

Let p_n/q_n be the continued-fraction convergents of β. The standard upper bound

$$\left|\beta - \frac{p_n}{q_n}\right| < \frac{1}{q_n q_{n+1}}$$

combined with (11) gives the following.

Corollary 4.2 (Polynomial denominator growth)**.** *There is an effective constant $C_M > 0$ such that*

$$q_{n+1} \leq C_M q_n^{\mu_M - 1} \tag{12}$$

for every n with $q_n \geq 2$. Consequently the next partial quotient satisfies $a_{n+1} = O_M(q_n^{\mu_M - 2})$.

This is qualitatively stronger than the inherited elementary estimate $q_{n+1} < 2M^{q_n}$. It rules out Liouville behavior and says that the exceptionally long near-repetitions of the Sturmian word are polynomially controlled in the preceding standard word length. The exponent depends on the unknown integer M, however, and may be enormous.

4.3 Polynomial spacing in conditional blocks

The irrationality measure also upgrades the separation of the block nodes. From (11), for $d \ge 1$ and the nearest integer r to $d\beta$,

$$\|d\beta\|_{\mathbb{Z}} = |d\beta - r| \ge c_M d^{1-\mu_M}. \tag{13}$$

For an integer $T \ge 0$, put

$$K_T = \left\lfloor \frac{T+1}{\beta} \right\rfloor, \qquad x_{T,k} = T - \lfloor k\beta \rfloor + k\beta = T + \{k\beta\}, \quad 0 \le k \le K_T.$$

The associated candidate and output nodes are

$$m_{T,k} = 2^{x_{T,k}} = 2^{T-\lfloor k\beta \rfloor} M^k, \qquad a_{T,k} = 3^{x_{T,k}} = 3^{T-\lfloor k\beta \rfloor} A^k. \tag{14}$$

Corollary 4.3 (Effective block separation)**.** *For distinct $i, j \in \{0, \dots, K_T\}$,*

$$|x_{T,j} - x_{T,i}| \ge c_M K_T^{1-\mu_M}, \tag{15}$$

$$|m_{T,j} - m_{T,i}| \ge (\log 2) c_M 2^T K_T^{1-\mu_M}, \tag{16}$$

$$|a_{T,j} - a_{T,i}| \ge (\log 3) c_M 3^T K_T^{1-\mu_M}. \tag{17}$$

Proof. The ordinary distance between two fractional parts is at least their circular distance, so

$$|\{j\beta\} - \{i\beta\}| \ge \|(j-i)\beta\|_{\mathbb{Z}} \ge c_M |j-i|^{1-\mu_M} \ge c_M K_T^{1-\mu_M},$$

because $1 - \mu_M < 0$. This proves (15). The mean value theorem applied to 2^x and 3^x on $[T, T+1]$ gives the remaining inequalities. ☐

4.4 Separation on the full (n, k) lattice

The same estimate applies to the two-dimensional conditional exponent grid.

Corollary 4.4 (Polynomial separation of conditional exponents)**.** *There are effective $c_M, \mu_M > 0$ such that, for distinct integer pairs $(n, k), (n', k')$ with $0 \le k, k' \le H$,*

$$|(n + k\beta) - (n' + k'\beta)| \ge \min\{1, c_M H^{1-\mu_M}\}.$$

Proof. If $k = k'$, the difference is a nonzero integer. Otherwise divide the linear form $(n-n') + (k-k')\beta$ by $|k - k'|$ and apply (11). ☐

The column exponent lattice $u + \vartheta v$ has an analogous effective polynomial spacing with absolute constants, because $\vartheta = \log 3 / \log 2$ is a fixed ratio of two logarithms. More refined explicit estimates are known for linear forms in $\log 2$ and $\log 3$; for example Wu's simultaneous linear-independence measure leads, in the usual normalization, to an irrationality exponent for ϑ below approximately 8.62 [8]. The exact convention-dependent numerical conversion is not needed below.

4.5 Why the Baker upgrade does not close the conjecture

The effective spacing is useful, but its scale is still compatible with the exact conditional population. A dyadic exponent block contains only $K_T + 1 \asymp T$ nodes. A polynomial lower bound on their minimum spacing imposes no contradiction on T points in a fixed interval. Likewise, the full triangular grid has $\asymp T^2$ points in an interval of exponent length T, and polynomially small gaps are expected.

Baker's theorem therefore strengthens compactness and determinant estimates but does not supply the missing logarithm by itself. Its most promising roles are indirect:

- controlling the smallest row and column gaps in an exact Harish–Chandra–Itzykson–Zuber factorization;
- bounding how often a p-adic assignment problem can be nearly degenerate;
- limiting the lengths of near-periodic Sturmian blocks that a congruence argument must handle;
- making computational searches over continued-fraction branches provably finite once a candidate output M is fixed.

5 Exact all-support valuations of ordinary block Vandermondes

5.1 The block determinants

Fix an integer T and abbreviate $K = K_T$. Let $m_k = m_{T,k}$ and $a_k = a_{T,k}$ be the $K+1$ input–output pairs in (14). Their ordinary Vandermonde determinants are

$$V_T^{(2)} = \det(m_i^j)_{0\le i,j\le K} = \prod_{0\le i<j\le K} (m_j - m_i), \tag{18}$$

$$V_T^{(3)} = \det(a_i^j)_{0\le i,j\le K} = \prod_{0\le i<j\le K} (a_j - a_i). \tag{19}$$

They are nonzero integers. Since $m_k \in [2^T, 2^{T+1})$ and $a_k \in [3^T, 3^{T+1})$,

$$\log |V_T^{(2)}| \le R_T T \log 2, \tag{20}$$

$$\log |V_T^{(3)}| \le R_T (T \log 3 + \log 2), \tag{21}$$

where

$$R_T = \binom{K+1}{2} = \frac{K(K+1)}{2}.$$

The additive $R_T \log 2$ in (21) is lower order.

5.2 The input determinant: an exact formula at every support prime

Factor

$$M = 2^a W, \qquad W \text{ odd}, \qquad \delta_2 = \beta - a = \log_2 W.$$

As noted above, $W > 1$, so $W \ge 3$ and

$$\delta_2 \ge \log_2 3 > 1. \tag{22}$$

Because a is an integer,

$$m_k = 2^{T-\lfloor k\delta_2 \rfloor} W^k. \tag{23}$$

The 2-adic valuations in (23) strictly decrease with k, while every odd support-prime valuation strictly increases.

Proposition 5.1 (Exact support valuations for $V_T^{(2)}$)**.** *For $0 \le i < j \le K$,*

$$v_2(m_j - m_i) = T - \lfloor j\delta_2 \rfloor, \tag{24}$$

$$v_p(m_j - m_i) = i\, v_p(W) \qquad (p \mid W). \tag{25}$$

Consequently

$$v_2(V_T^{(2)}) = \sum_{j=1}^{K} j(T - \lfloor j\delta_2 \rfloor), \tag{26}$$

$$v_p(V_T^{(2)}) = v_p(W) \sum_{i=0}^{K-1} i(K-i) \qquad (p \mid W). \tag{27}$$

Proof. Put $\rho_k = T - \lfloor k\delta_2 \rfloor$. By (22), $\rho_i > \rho_j$ when $i < j$. From (23),

$$m_j - m_i = 2^{\rho_j} W^i \left(W^{j-i} - 2^{\rho_i - \rho_j} \right).$$

The parenthesis is odd, proving (24). For an odd prime $p \mid W$, the two terms m_i and m_j have distinct valuations $iv_p(W) < jv_p(W)$, so the ultrametric equality gives (25). Summing over pairs gives (26)–(27). □

Let $Q_T^{(2)}$ be the part of $V_T^{(2)}$ supported on the primes dividing $2M$. Then Proposition 5.1 gives the exact identity

$$\log Q_T^{(2)} = \log 2 \sum_{j=1}^{K} j(T - \lfloor j\delta_2 \rfloor) + \log W \sum_{i=0}^{K-1} i(K-i). \tag{28}$$

The two elementary sums have asymptotics

$$\sum_{j=1}^{K} j(T - \lfloor j\delta_2 \rfloor) = \frac{TK^2}{2} - \frac{\delta_2 K^3}{3} + O_\beta(T^2),$$

$$\sum_{i=0}^{K-1} i(K-i) = \frac{K^3}{6} + O(K^2).$$

Using $\log W = \delta_2 \log 2$ and $K/T \to 1/\beta$ yields

$$\frac{\log Q_T^{(2)}}{R_T T \log 2} \longrightarrow 1 - \frac{\delta_2}{3\beta}. \tag{29}$$

This includes every prime appearing in the entries. Any further prime divisor of the Vandermonde comes from subtraction and hence from genuine cancellation beyond termwise entry valuations.

5.3 The output determinant and bounded tie corrections

Factor

$$A = 3^b Z, \qquad 3 \nmid Z, \qquad \delta_3 = \beta - b = \log_3 Z > 0.$$

Then

$$a_k = 3^{T - \lfloor k\delta_3 \rfloor} Z^k. \tag{30}$$

Unlike δ_2, the number δ_3 can be smaller than 1; for example the least possible base-free factor is 2. Equal consecutive 3-adic valuations can therefore occur. They contribute only a lower-order correction.

Proposition 5.2 (Output support valuations)**.** *For every prime $p \mid Z$ and $0 \le i < j \le K$,*

$$v_p(a_j - a_i) = i\, v_p(Z).$$

At $p = 3$,

$$v_3(a_j - a_i) = T - \lfloor j\delta_3 \rfloor$$

after adding a nonnegative correction that is zero whenever $\lfloor i\delta_3 \rfloor < \lfloor j\delta_3 \rfloor$. The total correction over all pairs is $O_{A,\beta}(K^2)$. Consequently, if $Q_T^{(3)}$ denotes the part of $V_T^{(3)}$ supported on primes dividing $3A$, then

$$\frac{\log Q_T^{(3)}}{R_T T \log 3} \longrightarrow 1 - \frac{\delta_3}{3\beta}. \tag{31}$$

Proof. The assertion for $p \mid Z$ is the same distinct-valuation argument as before. Put $\rho_k = T - \lfloor k\delta_3 \rfloor$. If $\rho_i > \rho_j$, then

$$a_j - a_i = 3^{\rho_j} Z^i \left(Z^{j-i} - 3^{\rho_i - \rho_j} \right)$$

and the parenthesis is not divisible by 3, so the valuation is exactly ρ_j. If $\rho_i = \rho_j$, then

$$v_3(a_j - a_i) = \rho_j + v_3(Z^{j-i} - 1).$$

Equality of the floors implies $(j-i)\delta_3 < 1$, so $j - i$ belongs to a fixed finite set depending only on δ_3. Hence the extra valuation is bounded by a constant depending on A and β. There are $O(K^2)$ pairs. The asymptotic calculation is then identical to (28). □

5.4 The depth-sensitive product ceiling

Combining (20), (21), (29), and (31) gives the principal conclusion of the ordinary Vandermonde audit.

Theorem 5.3 (All-support ordinary Vandermonde fraction)**.** *For a hypothetical primitive counterexample,*

$$\frac{\log(Q_T^{(2)} Q_T^{(3)})}{R_T T \log 6} \longrightarrow 1 - \frac{\delta_2 \log 2 + \delta_3 \log 3}{3\beta \log 6}, \qquad \delta_2 = \beta - a, \quad \delta_3 = \beta - b. \tag{32}$$

Because $\min(a, b) = 0$*, the limit is at most*

$$1 - \frac{\log 2}{3 \log 6} = 0.871049\,064\ldots. \tag{33}$$

If $b = 0$, the sharper ceiling is

$$1 - \frac{\log 3}{3 \log 6} = 0.795599\,921\ldots.$$

If $a = b = 0$, the limit is exactly $2/3$.

Proof. Only the numerical ceiling remains to be shown. If $a = 0$, then $\delta_2 = \beta$, so the deficit in (32) is at least $\log 2/(3\log 6)$. If $b = 0$, it is at least $\log 3/(3\log 6)$. Taking the larger possible fraction gives (33). If both vanish, the numerator of the deficit is $\beta \log 6$. □

Table 2: Leading termwise-divisor ceilings for ordinary block Vandermondes.

Branch information	Exact/upper fraction	Decimal
$a = b = 0$	$2/3$	$0.666666\ldots$
$b = 0$	$\leq 1 - \log 3/(3\log 6)$	$0.795599\ldots$
No matched depth only	$\leq 1 - \log 2/(3\log 6)$	$0.871049\ldots$

5.5 Why even 87% is not enough

The deficit in Theorem 5.3 is a fixed positive fraction once the hypothetical primitive pair is fixed. The archimedean height is $\asymp T^3$, whereas all factorial, spacing, and tie corrections are $O_\beta(T^2 \log T)$. Thus no lower-order refinement of the ordinary Vandermonde can close the remaining gap.

This conclusion is stronger than the earlier observation that the 2-adic valuation alone captures one third in the base-free case. It includes every prime already present in the entries and every possible base depth. A successful ordinary-Vandermonde proof would need a new leading-order source of divisibility in the prime factors of the *differences*; the fixed-support valuations are completely accounted for by Theorem 5.3.

6 The mixed-semigroup determinant and its Gini norm

6.1 Balanced low-weight columns

Let

$$\Lambda = \{u + \vartheta v : (u, v) \in \mathbb{N}_0^2\}.$$

Because ϑ is irrational, every element has a unique pair (u, v). List the pairs in increasing order of $u + \vartheta v$:

$$(u_0, v_0), (u_1, v_1), \ldots.$$

For $N \geq 1$, let

$$L_N = \max_{0 \leq j < N} (u_j + \vartheta v_j).$$

The elementary lattice count

$$\#\{(u, v) \in \mathbb{N}_0^2 : u + \vartheta v \leq L\} = \frac{L^2}{2\vartheta} + O_\vartheta(L) \tag{34}$$

shows that

$$L_N \sim \sqrt{2\vartheta N}. \tag{35}$$

Moreover, the empirical distribution of

$$X_j = \left(\frac{u_j}{L_N}, \frac{\vartheta v_j}{L_N}\right) \tag{36}$$

converges to uniform area measure on the standard triangle

$$\mathcal{T} = \{(x, z) \in \mathbb{R}^2 : x \geq 0, z \geq 0, x + z \leq 1\}. \tag{37}$$

Here and below "uniform" means probability measure with density 2 on this triangle.

For the block at height T, set $K = K_T$ and $N = K + 1$. Define

$$D_T = \det \left(m_{T,k}^{u_j} a_{T,k}^{v_j} \right)_{0 \leq k,j < N}. \tag{38}$$

The generalized-Vandermonde theorem inherited from the source report implies that D_T is a nonzero integer. Put

$$B_j = 2^{u_j} 3^{v_j}.$$

Since $m_{T,k} = 2^{x_{T,k}}$ and $a_{T,k} = 3^{x_{T,k}}$,

$$m_{T,k}^{u_j} a_{T,k}^{v_j} = B_j^{x_{T,k}} = B_j^{T+\{k\beta\}}. \tag{39}$$

Thus the Leibniz expansion gives the clean archimedean bound

$$\log |D_T| \leq (T + 1) \sum_{j<N} \log B_j + \log N!. \tag{40}$$

The leading height is

$$H_T = T \sum_{j<N} \log B_j. \tag{41}$$

By (35) and the mean $\mathbb{E}(X_1 + X_2) = 2/3$ on $\mathcal{T}$,

$$H_T \sim \frac{2}{3} T N L_N \log 2 \asymp_\beta T^{5/2}. \tag{42}$$

Both $\sum_j \log B_j$ and $\log N!$ in the error of (40) are $o(H_T)$.

6.2 The all-prime tropical divisor

For a prime p, define the assignment minimum

$$\tau_{p,T} = \min_{\sigma \in S_N} \sum_{k=0}^{N-1} v_p \left(m_{T,k}^{u_{\sigma(k)}} a_{T,k}^{v_{\sigma(k)}} \right). \tag{43}$$

Every Leibniz term is divisible by $p^{\tau_{p,T}}$, so

$$Q_T^{\text{trop}} = \prod_p p^{\tau_{p,T}} \quad \text{divides} \quad D_T. \tag{44}$$

Only primes dividing $6MA$ occur in the product.

The problem is now to compute the leading asymptotic of $\log Q_T^{\text{trop}} / H_T$. A crude estimate "minimum $\leq$ average" loses the precise branch dependence. The exact answer is controlled by a simple probability norm.

6.3 Antimonotone assignment and the Gini norm

Let $X = (X_1, X_2)$ be uniform on $\mathcal{T}$, and let X' be an independent copy. For $c = (r, s) \in \mathbb{R}^2$, define

$$\mathfrak{G}(r, s) = \mathbb{E}\left|r(X_1 - X_1') + s(X_2 - X_2')\right|. \tag{45}$$

This is a norm on $\mathbb{R}^2$: homogeneity and the triangle inequality are immediate, and the support of $X - X'$ spans $\mathbb{R}^2$, so only $(r, s) = (0, 0)$ has norm zero.

Let $Y = rX_1 + sX_2$, with ascending quantile $Q_Y(t)$. Pairing an increasing uniform row parameter with the decreasing ordering of Y gives the continuum assignment functional

$$\Phi(r, s) = \int_0^1 t\, Q_Y(1 - t)\, dt. \tag{46}$$

Lemma 6.1 (Gini formula for the minimum assignment)**.** *For every $(r, s) \in \mathbb{R}^2$,*

$$\Phi(r, s) = \frac{1}{2}\,\mathbb{E}Y - \frac{1}{4}\,\mathfrak{G}(r, s) = \frac{r + s}{6} - \frac{1}{4}\,\mathfrak{G}(r, s). \tag{47}$$

Proof. The rearrangement inequality says that the minimum scalar product pairs the increasing row parameter with the decreasing list of column values. Changing variables $u = 1 - t$ in (46) gives

$$\Phi = \int_0^1 (1 - u) Q_Y(u)\, du.$$

If Y' is an independent copy, the minimum of Y, Y' has quantile density $2(1 - u)$, so

$$\Phi = \frac{1}{2}\mathbb{E}\min(Y, Y').$$

Since $\min(Y, Y') = (Y + Y' - |Y - Y'|)/2$, this becomes $\frac{1}{2}\mathbb{E}Y - \frac{1}{4}\mathbb{E}|Y - Y'|$. Finally $\mathbb{E}X_1 = \mathbb{E}X_2 = 1/3$. □

6.4 Closed form of the Gini norm

The norm (45) is elementary. Sort the three vertex values

$$0, \qquad r, \qquad s$$

as $\xi_0 \le \xi_1 \le \xi_2$, and put

$$A_0 = \xi_1 - \xi_0, \qquad B_0 = \xi_2 - \xi_1.$$

Proposition 6.2 (Explicit triangular Gini norm)**.** *If $A_0 + B_0 > 0$, then*

$$\mathfrak{G}(r, s) = \frac{2\left(2A_0^2 + 3A_0B_0 + 2B_0^2\right)}{15(A_0 + B_0)}. \tag{48}$$

At $(r, s) = (0, 0)$ the value is 0. In particular,

$$\mathfrak{G}(1, 0) = \mathfrak{G}(0, 1) = \mathfrak{G}(1, 1) = \frac{4}{15}, \qquad \mathfrak{G}(1, -1) = \frac{7}{15}. \tag{49}$$

Proof. Translation of the three vertex values does not change a difference $Y - Y'$, so take $\xi_0 = 0$, $\xi_1 = A_0$, and $\xi_2 = A_0 + B_0 = L$. Projection of uniform area measure on a triangle by a linear function has density

$$f(y) = \begin{cases} \dfrac{2y}{A_0 L}, & 0 \le y \le A_0, \\ \dfrac{2(L-y)}{B_0 L}, & A_0 \le y \le L. \end{cases}$$

Thus its distribution function is $F(y) = y^2/(A_0 L)$ on the first interval and $F(y) = 1 - (L-y)^2/(B_0 L)$ on the second. The standard identity

$$\mathbb{E}|Y - Y'| = 2 \int_{-\infty}^{\infty} F(y)(1 - F(y))\, dy$$

gives (48) after two polynomial integrations. Cases with one gap zero follow by continuity. □

For later use, if $\gamma \ge 0$, then

$$\mathfrak{G}(-1, \gamma) = \frac{2(2 + 3\gamma + 2\gamma^2)}{15(1+\gamma)} = \frac{4}{15} + \frac{2\gamma(1 + 2\gamma)}{15(1+\gamma)} \ge \frac{4}{15}. \tag{50}$$

6.5 Normalized prime-valuation slope vectors

Recall

$$a = v_2(M), \quad b = v_3(A), \quad c = v_2(A), \quad d = v_3(M).$$

For each prime, define a normalized slope vector $c_p \in \mathbb{R}^2$ by

$$c_2 = \left(-1 + \frac{a}{\beta}, \frac{c}{\beta\vartheta}\right), \tag{51}$$

$$c_3 = \left(\frac{\vartheta d}{\beta}, -1 + \frac{b}{\beta}\right), \tag{52}$$

$$c_p = \left(\frac{v_p(M)\log p}{\beta \log 2}, \frac{v_p(A)\log p}{\beta \log 3}\right), \qquad p \ne 2, 3. \tag{53}$$

The logarithmic factorizations of M and A give the conservation law

$$\sum_p c_p = (0, 0). \tag{54}$$

Indeed, the first coordinate sum is

$$-1 + \frac{a \log 2 + d \log 3 + \sum_{p \ne 2,3} v_p(M) \log p}{\beta \log 2} = 0,$$

and similarly for the second coordinate.

The meaning of these vectors can be read directly from the valuation matrices. Put $s_k = \beta k/T$. Up to $o(L_N)$ after division by T, the logarithmic p-adic cost of the entry indexed by a scaled column (x, z) is

Prime	Cost divided by $TL_N \log 2$
2	$x + s_k\, c_2 \cdot (x, z)$
3	$z + s_k\, c_3 \cdot (x, z)$
$p \ne 2, 3$	$s_k\, c_p \cdot (x, z)$

For example,

$$v_2(m_k^u a_k^v) = u(T - \lfloor k\beta \rfloor + ak) + vck,$$

and substituting $u = L_N x$, $v = L_N z/\vartheta$, $k \sim Ts_k/\beta$ gives the first row. The $O(u)$ errors from the floors sum to $O(NL_N) = o(H_T)$.

6.6 The exact continuum limit

Theorem 6.3 (All-prime tropical fraction for the mixed block determinant)**.** *Under a hypothetical primitive counterexample, the determinant* (38) *satisfies*

$$\frac{\log Q_T^{\mathrm{trop}}}{H_T} \longrightarrow 1 - \frac{3}{8} \sum_p \mathfrak{G}(c_p), \tag{55}$$

where the finitely many slope vectors c_p are given by (51)*–*(53)*.*

Proof. The row parameters $s_k = \beta k/T$ become uniformly distributed on $[0, 1]$, because $K_T \sim T/\beta$. The scaled columns become uniformly distributed on $\mathcal{T}$ by (34). The finite rearrangement inequality and Riemann-sum convergence therefore show that the assignment minimum at prime p, after normalization by $TL_N N \log 2$, tends to its constant-column mean plus $\Phi(c_p)$.

Only $p = 2$ has constant part x, and only $p = 3$ has constant part z. Their combined mean is $2/3$. By Lemma 6.1, the sum of the slope contributions is

$$\sum_p \Phi(c_p) = \frac{1}{6} \sum_p (c_{p,1} + c_{p,2}) - \frac{1}{4} \sum_p \mathfrak{G}(c_p).$$

The first sum vanishes by (54). Hence

$$\frac{\log Q_T^{\mathrm{trop}}}{TL_N N \log 2} \longrightarrow \frac{2}{3} - \frac{1}{4} \sum_p \mathfrak{G}(c_p).$$

On the other hand, (42) gives $H_T/(TL_N N \log 2) \to 2/3$. Dividing proves (55). □

This theorem is exact at the level of minimum Leibniz-term valuations. It replaces a family of separate p-adic estimates by one finite norm sum satisfying a conservation law.

The four base and cross depths already give a useful branch-sensitive bound without knowing any external prime factorization.

Corollary 6.4 (Depth-sensitive three-vector bound)**.** *For every hypothetical primitive counterexample,*

$$\limsup_{T\to\infty} \frac{\log Q_T^{\mathrm{trop}}}{H_T} \le 1 - \frac{3}{8}\Big(\mathfrak{G}(c_2) + \mathfrak{G}(c_3) + \mathfrak{G}(-c_2 - c_3)\Big). \tag{56}$$

Equality in this upper bound can occur only when all nonzero external slope vectors are positively collinear.

Proof. The external vectors satisfy $\sum_{p\neq 2,3} c_p = -c_2 - c_3$. Hence the triangle inequality gives

$$\sum_{p\neq 2,3} \mathfrak{G}(c_p) \ge \mathfrak{G}(-c_2 - c_3).$$

Substitute this in Theorem 6.3. Equality in a norm triangle inequality requires all summands to lie on one nonnegative ray. □

The equality condition detects exactly the second-iterate kernel branch reviewed in the source report. Indeed the external vectors c_p are rescaled versions of $(v_p(M), v_p(A))$. They are all collinear if and only if the two external valuation vectors of M and A are rationally dependent, equivalently if the kernel K_β is nonzero. Thus:

Corollary 6.5 (The tropical norm sees the kernel dichotomy)**.** *In the multiplicatively independent branch $K_\beta = 0$, the inequality in (56) is strict. In the dependent branch $K_\beta \neq 0$, the external triangle inequality is an equality (including the axis cases in which one output is $\{2,3\}$-smooth). For fixed M, A, the strict gap in the independent branch is explicit:*

$$\Delta_{\text{ext}} = \sum_{p \neq 2,3} \mathfrak{G}(c_p) - \mathfrak{G}\left(\sum_{p \neq 2,3} c_p\right) > 0. \tag{57}$$

There is no uniform positive lower bound for Δ_{ext}: distinct rational rays can approach one another as the valuations grow. Nevertheless, the quantity is a computable branch invariant and gives a sharper determinant threshold for every fixed hypothetical primitive pair.

6.7 The universal 4/5 ceiling

Theorem 6.6 (Sharp universal min-plus ceiling)**.** *For every hypothetical primitive counterexample,*

$$\sum_p \mathfrak{G}(c_p) \geq \frac{8}{15}. \tag{58}$$

Consequently

$$\limsup_{T \to \infty} \frac{\log Q_T^{\text{trop}}}{H_T} \leq \frac{4}{5}. \tag{59}$$

The constant 4/5 is the sharp supremum allowed by the normalized valuation constraints, although equality would require a degenerate boundary incompatible with a genuine nonintegral solution.

Proof. By the primitive theorem, either $a = 0$ or $b = 0$. Suppose first that $a = 0$. Then $c_2 = (-1, \gamma)$ with $\gamma = c/(\beta\vartheta) \geq 0$. By (50), $\mathfrak{G}(c_2) \geq 4/15$. Conservation and the triangle inequality for the norm give

$$\sum_{p \neq 2} \mathfrak{G}(c_p) \geq \mathfrak{G}\left(\sum_{p \neq 2} c_p\right) = \mathfrak{G}(-c_2) = \mathfrak{G}(c_2).$$

Thus the total is at least 8/15. If $b = 0$, apply the same argument to $c_3 = (\eta, -1)$, where $\eta = \vartheta d/\beta \geq 0$. Substitution in (55) gives $1 - (3/8)(8/15) = 4/5$. □

The proof is short because the difficult arithmetic has been compressed into two facts: the slope vectors sum to zero, and one structural vector reaches from -1 to a nonnegative vertex. No averaging estimate alone reveals this geometry.

6.8 Cross-coprime constants: 7/10 and 3/10

Assume the stronger branch condition

$$\gcd(MA, 6) = 1. \tag{60}$$

Then $a = b = c = d = 0$,

$$c_2 = (-1, 0), \qquad c_3 = (0, -1), \qquad \sum_{p \neq 2,3} c_p = (1, 1).$$

Corollary 6.7 (All-prime cross-coprime ceiling)**.** *Under* (60)*,*

$$\limsup_{T\to\infty} \frac{\log Q_T^{\mathrm{trop}}}{H_T} \leq \frac{7}{10}. \tag{61}$$

If the external normalized valuation vectors are not all positively collinear, the inequality is strict by a branch-dependent positive amount.

Proof. By (49), the structural vectors each contribute $4/15$. The triangle inequality gives

$$\sum_{p\neq 2,3} \mathfrak{G}(c_p) \geq \mathfrak{G}(1,1) = \frac{4}{15}.$$

Thus the total norm is at least $4/5$, and $1 - (3/8)(4/5) = 7/10$. Equality in the triangle inequality requires positive collinearity. $\square$

The places 2 and 3 alone have an even smaller exact continuum contribution.

Corollary 6.8 (The structural-place 3/10 constant)**.** *Under* (60)*, let* $Q_T^{(2,3)} = 2^{\tau_{2,T}} 3^{\tau_{3,T}}$*. Then*

$$\frac{\log Q_T^{(2,3)}}{H_T} \longrightarrow \frac{3}{10}. \tag{62}$$

Proof. For $c = (-1, 0)$,

$$\Phi(c) = \frac{-1}{6} - \frac{1}{4} \cdot \frac{4}{15} = -\frac{7}{30},$$

and the same holds for $(0, -1)$. Adding the constant mean $2/3$ gives $2/3 - 7/15 = 1/5$. Dividing by the height mean $2/3$ gives $3/10$. $\square$

Equivalently, one can compute this constant from quantiles. The descending quantile of X_1 on $\mathcal{T}$ is $1 - \sqrt{t}$, so

$$\int_0^1 t(1-\sqrt{t})\,dt = \frac{1}{10}, \qquad \mathbb{E}X_1 = \frac{1}{3}.$$

The ratio is $(1/10)/(1/3) = 3/10$ in each structural coordinate.

6.9 Correction and calibration of a later repository note

A short p-adic priority note added to the repository after the attached unified report makes the correct strategic observation that minimum Leibniz-term valuations leave a constant fraction of the archimedean height uncovered. Two details require correction.

First, the displayed structural valuation formulas in that note simultaneously omit the terms $kuv_2(M)$ and $kvv_3(A)$; they therefore assume both $a = 0$ and $b = 0$. The inherited primitive theorem gives only $\min(a, b) = 0$. Equations (51)–(52) retain all four cross/base depths.

Second, the proposed all-prime 2/3 ceiling is not a valid universal consequence of "minimum $\leq$ average." For an external positive weight proportional to $X_1 + X_2$, the minimum assignment is 4/5 of its average assignment, not at most 2/3. The exact Gini formula shows that the universal normalized ceiling is 4/5, while the cross-coprime all-prime ceiling is 7/10. The note's qualitative conclusion survives—entrywise valuation is insufficient—but the constants and branch hypotheses are materially different.

Status. [new paper proof]

The correction is not based on a numerical counterexample to the Alaoglu–Erdős conjecture. It is an exact audit of the assignment inequality used in the method. Rational external valuation directions can approximate the equality configurations arbitrarily well, so no stronger uniform constant follows merely from integrality of the valuation vectors.

6.10 Where extra divisibility can occur

The tropical minimum equals the true determinant valuation whenever its minimizing permutation is unique.

Proposition 6.9 (Unique tropical term forbids cancellation)**.** *Let $D = \det(E_{ij})$ be a nonzero integer determinant. Fix a prime p, and suppose a unique permutation σ_0 minimizes $\sum_i v_p(E_{i,\sigma(i)})$. Then*

$$v_p(D) = \sum_i v_p(E_{i,\sigma_0(i)}).$$

Proof. Divide the Leibniz expansion by the corresponding power of p. The term indexed by σ_0 is a p-adic unit, while every other term is divisible by p. The sum is therefore a unit modulo p. □

Consequently, the missing 20% or more can arise only from primes at which the assignment problem has several minimizers. The tie geometry is explicit.

- For $p \neq 2, 3$, columns tie when $v_p(M)(u - u') + v_p(A)(v - v') = 0$, so ties lie on rational lines orthogonal to the external valuation vector.
- In the cross-coprime branch, the 2-adic cost depends only on u and the 3-adic cost only on v. Every vertical or horizontal column fiber creates many equal-cost permutations. Any leading extra valuation must be produced inside these fibers.
- Cross valuations $v_2(A)$ and $v_3(M)$ tilt the fibers. The floor term in the row profile perturbs the exact affine ordering but does not change the continuum slope vector.

This turns "look for p-adic cancellation" into a concrete algebra problem: factor or control the residual determinants inside equal-slope column fibers, and show that their valuation supplies at least the missing Gini deficit.

7 The complete triangular grid and a $\pi/4$ transport ceiling

The one-block determinant deliberately uses the exact $O(T)$ population in a unit interval of exponent space. The source report also suggests exploiting the full two-dimensional coordinate grid rather than sorting only by magnitude. Conditional on a counterexample, that proposal gives $\asymp T^2$ genuine integer rows below exponent height T, exactly the number that the mixed-semigroup interpolation argument seemed to be missing. It is therefore important to determine whether entrywise p-adic divisibility becomes strong enough on this larger node set.

The answer is again negative, but for a different and pleasantly universal reason. Once all places are summed, the prime-by-prime assignment minima are bounded by one antimonotone transport problem for the logarithmic entry matrix. Both the row heights and the balanced column weights have the same triangular distribution, and the ratio between antimonotone and monotone transport is exactly $\pi/4$.

7.1 Rows and columns

For $T > 0$ define the complete conditional row set

$$\mathcal{R}_T = \{(n, k) \in \mathbb{N}_0^2 : n + k\beta \le T\}, \qquad N_T = \#\mathcal{R}_T. \tag{63}$$

For $i = (n_i, k_i) \in \mathcal{R}_T$ put

$$x_i = n_i + k_i\beta, \qquad m_i = 2^{n_i} M^{k_i} = 2^{x_i}, \qquad a_i = 3^{n_i} A^{k_i} = 3^{x_i}. \tag{64}$$

Because β is irrational, the x_i are pairwise distinct.

Let $\mathcal{C}_T$ be the first N_T points $(u, v) \in \mathbb{N}_0^2$ in increasing order of

$$\lambda = u + \vartheta v, \qquad y = \log(2^u 3^v) = \lambda \log 2.$$

Write L_T for the largest selected λ. Standard lattice-point estimates for the two triangles give

$$N_T = \frac{T^2}{2\beta} + O_\beta(T), \qquad L_T = \sqrt{2\vartheta N_T} + O_\vartheta(1) = T\sqrt{\frac{\vartheta}{\beta}} + O_\beta(1). \tag{65}$$

The associated square determinant is

$$D_T^{\triangle} = \det\Big(m_i^{u_j} a_i^{v_j}\Big)_{i,j=1}^{N_T} = \det(e^{x_i y_j})_{i,j=1}^{N_T}. \tag{66}$$

Every entry is an integer. Ordering the x_i and y_j increasingly, strict total positivity of the exponential kernel makes $D_T^{\triangle} > 0$; equivalently, this is the same exponential-polynomial zero theorem used by the generalized-Vandermonde part of the source report.

Status. [new paper proof] for the asymptotic transport statements below. Integrality and nonvanishing are inherited from the mixed-semigroup determinant architecture [Lean].

7.2 The common triangular distribution

The normalized row heights $h_i = x_i/T$ and column weights $\ell_j = y_j/(L_T \log 2)$ both live in $[0, 1]$. Their empirical measures have the same limit.

Lemma 7.1 (Triangular height law)**.** *For every continuous $f : [0,1] \to \mathbb{R}$,*

$$\frac{1}{N_T}\sum_{i=1}^{N_T} f(h_i) \longrightarrow \int_0^1 2tf(t)\,dt, \tag{67}$$

$$\frac{1}{N_T}\sum_{j=1}^{N_T} f(\ell_j) \longrightarrow \int_0^1 2tf(t)\,dt. \tag{68}$$

Thus the common limiting distribution has cumulative distribution function $F(t) = t^2$ and quantile function

$$Q(s) = \sqrt{s}, \qquad 0 \le s \le 1. \tag{69}$$

Proof. After scaling (n,k) to $(n/T, \beta k/T)$, the row lattice becomes equidistributed in the unit right triangle

$$\{(r,s) : r,s \ge 0,\ r+s \le 1\}.$$

The subtriangle on which $r + s \le t$ has area $t^2/2$, while the whole triangle has area $1/2$. Hence the height $r + s$ has distribution function t^2. Boundary discrepancies are $O(T)$ against $N_T \asymp T^2$, proving (67).

For the columns, scale (u,v) to $(u/L_T, \vartheta v/L_T)$. The same unit triangle appears, and $\ell = (u+\vartheta v)/L_T$ is again the sum of the scaled coordinates. The boundary shell created by taking the first exactly N_T points has $O(L_T)$ points against $N_T \asymp L_T^2$, which proves (68). □

7.3 Summing all primewise assignment minima

For each prime p, let

$$\tau_{p,T} = \min_{\sigma\in S_{N_T}} \sum_{i=1}^{N_T} v_p\Big(m_i^{u_{\sigma(i)}} a_i^{v_{\sigma(i)}}\Big), \qquad Q_T^{\triangle} = \prod_p p^{\tau_{p,T}}. \tag{70}$$

As before, $Q_T^{\triangle}$ divides every Leibniz term and hence divides $D_T^{\triangle}$. The key observation is that taking a separate minimizing permutation at each prime can only decrease the sum relative to using one permutation at all primes.

Lemma 7.2 (All-place transport domination)**.** *For every T,*

$$\log Q_T^{\triangle} \le \min_{\sigma\in S_{N_T}} \sum_{i=1}^{N_T} x_i y_{\sigma(i)}. \tag{71}$$

Proof. For a permutation σ, put

$$C_p(\sigma) = \sum_i v_p\Big(m_i^{u_{\sigma(i)}} a_i^{v_{\sigma(i)}}\Big)\log p.$$

Then

$$\sum_p \min_\sigma C_p(\sigma) \le \min_\sigma \sum_p C_p(\sigma).$$

Unique factorization and (64) give, entry by entry,

$$\sum_p v_p(m_i^{u_j} a_i^{v_j})\log p = \log(m_i^{u_j} a_i^{v_j}) = x_i y_j.$$

Substitution proves (71). □

Notice how little arithmetic remains in the last inequality: it is valid in every valuation branch, includes all primes, and does not use the primitive-depth normalization. The price is that it is only an upper bound for the common termwise divisor. That is exactly what is needed for a no-go theorem.

7.4 Optimal transport and the $\pi/4$ ratio

Let

$$E_T^- = \min_{\sigma \in S_{N_T}} \sum_i x_i y_{\sigma(i)}, \tag{72}$$

$$E_T^+ = \max_{\sigma \in S_{N_T}} \sum_i x_i y_{\sigma(i)}. \tag{73}$$

The rearrangement inequality pairs the two ordered lists oppositely for E_T^- and in the same order for E_T^+.

Theorem 7.3 (Triangular-grid transport constants). *As $T \to \infty$,*

$$\frac{E_T^-}{N_T T L_T \log 2} \longrightarrow \int_0^1 \sqrt{t(1-t)}\,dt = \frac{\pi}{8}, \tag{74}$$

$$\frac{E_T^+}{N_T T L_T \log 2} \longrightarrow \int_0^1 t\,dt = \frac{1}{2}. \tag{75}$$

Consequently

$$\frac{E_T^-}{E_T^+} \longrightarrow \frac{\pi}{4} = 0.7853981633\ldots. \tag{76}$$

Proof. By Lemma 7.1, the increasing empirical quantile functions of the two normalized lists converge in L^1 to $Q(t) = \sqrt{t}$. The discrete rearrangement sums are Riemann sums for the monotone and antimonotone couplings. Thus

$$\frac{E_T^-}{N_T T L_T \log 2} \longrightarrow \int_0^1 Q(t)Q(1-t)\,dt, \qquad \frac{E_T^+}{N_T T L_T \log 2} \longrightarrow \int_0^1 Q(t)^2\,dt.$$

The first integral is the beta integral $B(3/2, 3/2) = \Gamma(3/2)^2/\Gamma(3) = \pi/8$; the second is $1/2$. □

The determinant has $N_T!$ Leibniz terms, each at most $e^{E_T^+}$. Since $N_T \asymp T^2$ while $T L_T N_T \asymp T^4$,

$$\log(N_T!) = O(T^2 \log T) = o(N_T T L_T). \tag{77}$$

Combining the preceding statements gives the promised ceiling.

Corollary 7.4 (Full-grid termwise-divisor ceiling). *With the notation above,*

$$\limsup_{T\to\infty} \frac{\log Q_T^{\triangle}}{E_T^+ + \log(N_T!)} \le \frac{\pi}{4}. \tag{78}$$

Thus the complete $\asymp T^2$ row grid does not make the all-prime common divisor exceed the sharp largest-Leibniz-term archimedean bound. Entrywise valuation alone leaves at least the fraction $1 - \pi/4 = 0.2146018\ldots$ uncovered at leading order.

Proof. Lemma 7.2 gives $\log Q_T^{\triangle} \le E_T^-$. Divide by the standard determinant upper bound $E_T^+ + \log(N_T!)$ and apply Theorem 7.3 and (77). □

Caution 7.5 (What the $\pi/4$ theorem does not exclude). The denominator in (78) is the largest-term upper bound, not the true size of $D_T^{\triangle}$. Generalized Vandermonde determinants exhibit substantial alternating cancellation. A sufficiently sharp archimedean estimate could lower the relevant budget below E_T^-, in which case the termwise divisor might become decisive. The theorem excludes the naive "more rows plus the same Hadamard/Leibniz estimate" route; it does not exclude a joint p-adic/free-energy argument.

7.5 The Harish–Chandra–Itzykson–Zuber identity

The exact analytic object behind that warning is classical. For diagonal Hermitian matrices $X = \operatorname{diag}(x_1, \dots, x_N)$ and $Y = \operatorname{diag}(y_1, \dots, y_N)$ with distinct increasing entries, normalized Haar measure on the unitary group satisfies

$$\int_{U(N)} \exp\big(\operatorname{Tr}(XUYU^*)\big)\, dU = \frac{\prod_{r=1}^{N-1} r!}{\Delta(x)\Delta(y)} \det(e^{x_i y_j})_{i,j=1}^N, \tag{79}$$

where $\Delta(x) = \prod_{i<j}(x_j - x_i)$ and similarly for $\Delta(y)$ [13, 14, 15]. Hence

$$\log D_T^{\triangle} = \log \Delta(x) + \log \Delta(y) - \sum_{r=1}^{N_T - 1} \log(r!) + \log \mathcal{I}_T, \tag{80}$$

with $\mathcal{I}_T$ the unitary integral in (79).

The large powers of T hidden separately in the two Vandermonde factors and the factorial product cancel at leading logarithmic scale. What remains is a genuine free-energy problem for two triangular empirical measures. The Baker separation theorem of Section 4 gives a polynomial lower bound for the spacings in $\Delta(x)$, so the row Vandermonde cannot collapse through anomalously close conditional exponents. What is not known is a bound on the full combination (80) strong enough to beat the primewise divisor.

Research target 7.6 (HCIZ–valuation comparison)**.** Determine the T^4-scale limit, or a sufficiently sharp upper bound, of

$$\log D_T^{\triangle} - \log Q_T^{\triangle}$$

for the conditional triangular row lattice and balanced mixed columns. It would suffice to prove that this quantity is negative for all sufficiently large T. Corollary 7.4 shows that such a result must use either the HCIZ/Vandermonde cancellation in (80), extra p-adic cancellation beyond the assignment minima, or both.

This formulation has a useful advantage over a generic interpolation determinant: both limiting measures and all lattice densities are explicit. It is therefore suitable for numerical free-energy reconnaissance before attempting a proof.

8 The first tied p-adic layer is lower order

Theorem 6.6 says that extra divisibility must come from ties among minimum-valuation Leibniz terms. Ties are especially abundant in the cross-coprime branch (60): at $p = 2$ all columns with the same u

have the same valuation profile, and at $p = 3$ all columns with the same v do. It is natural to hope that cancellation inside those large fibers supplies the missing leading-order fraction.

The present section computes that first cancellation layer exactly. It factors into ordinary Vandermondes of the odd row parts. Standard upper bounds for p-adic linear forms in two logarithms then show that the whole first layer is only $O(T^{3/2}(\log T)^C)$, whereas the mixed determinant has height $\asymp T^{5/2}$. This does not bound the final determinant valuation—higher tropical layers may cancel again—but it rules out a one-step fiber explanation of the missing fraction.

8.1 Tropical initial form at two

Assume throughout this section that $\gcd(MA, 6) = 1$. For the block rows, write

$$q_k = a_{T,k} = 3^{n_k} A^k, \qquad n_k = T - \lfloor k\beta \rfloor. \tag{81}$$

Both M and every q_k are odd, and

$$m^u_{T,k} a^v_{T,k} = 2^{un_k}\, M^{ku} q^v_k. \tag{82}$$

For each integer $u \geq 0$, let

$$V_u = \{v : (u, v) \text{ is one of the selected } N \text{ columns}\}, \qquad s_u = \#V_u.$$

Because columns are selected by the inequality $u + \vartheta v \leq L_N$ up to one boundary shell, each nonempty V_u is an initial interval of integers. Sort the distinct u values in increasing order. Since $n_0 > n_1 > \cdots > n_{N-1}$, every minimizing assignment for the cost un_k sends one consecutive row block R_u of size s_u to the columns in the fiber $\{u\} \times V_u$.

Let $\tau_{2,T}$ be the minimum from (43). The sum of all Leibniz terms having exactly this valuation is the 2-adic tropical initial form. Up to an overall sign and an odd factor, it is

$$I_{2,T} = \prod_u \det(q^v_k)_{k\in R_u,\ v\in V_u} = \prod_u \prod_{\substack{i<j \\ i,j\in R_u}} (q_j - q_i). \tag{83}$$

The first equality follows by grouping the optimal permutations by fibers and factoring the odd number M^{ku} from each row in a fixed u block; the second is the ordinary Vandermonde identity because V_u is consecutive.

Proposition 8.1 (Exact first 2-adic initial form)**.** *There is an integer $Z_{2,T}$ and an integer gap $g_{2,T} \geq \lfloor \beta \rfloor$ such that*

$$2^{-\tau_{2,T}} D_T = \varepsilon_{2,T} I_{2,T} + 2^{g_{2,T}} Z_{2,T}, \tag{84}$$

where $\varepsilon_{2,T}$ is odd. Consequently

$$v_2(D_T) - \tau_{2,T} \geq \min\{v_2(I_{2,T}), g_{2,T}\}, \tag{85}$$

and equality holds whenever the two numbers inside the minimum are unequal.

Proof. The rearrangement inequality with repeated column weights identifies the minimizing permutations exactly as above, and summing their signed unit parts gives (83). Any nonminimizing assignment must move at least one column across two adjacent u fibers. An exchange across the corresponding boundary changes the cost by

$$(u' - u)(n_i - n_j).$$

Here $u' - u \geq 1$, while consecutive row depths differ by $\lfloor (k + 1)\beta \rfloor - \lfloor k\beta \rfloor \geq \lfloor \beta \rfloor$. Hence every nonminimal Leibniz term has valuation at least $\tau_{2,T} + \lfloor \beta \rfloor$, proving the expansion. The valuation statement follows from the ultrametric inequality. □

8.2 The symmetric initial form at three

Put

$$r_k = m_{T,k} = 2^{n_k} M^k. \tag{86}$$

At $p = 3$ the entry factors as

$$m_{T,k}^u a_{T,k}^v = 3^{vn_k} r_k^u A^{kv}.$$

For a selected value of v, let U_v be the consecutive set of selected u values and let R'_v be the corresponding consecutive block of rows in the minimum assignment. The same argument gives

$$I_{3,T} = \prod_v \prod_{\substack{i<j \\ i,j \in R'_v}} (r_j - r_i) \tag{87}$$

and

$$3^{-\tau_{3,T}} D_T = \varepsilon_{3,T} I_{3,T} + 3^{g_{3,T}} Z_{3,T}, \qquad g_{3,T} \ge \lfloor \beta \rfloor, \tag{88}$$

with $\varepsilon_{3,T}$ a 3-adic unit.

8.3 Two-logarithm bounds inside the fibers

For $i < j$, set $d = j - i$ and $h = n_i - n_j = \lfloor j\beta \rfloor - \lfloor i\beta \rfloor$. The common factors in the row values are units at the relevant place, so

$$v_2(q_j - q_i) = v_2(A^d - 3^h), \tag{89}$$

$$v_3(r_j - r_i) = v_3(M^d - 2^h). \tag{90}$$

Neither difference vanishes: $A^d = 3^h$ or $M^d = 2^h$ would give $d\beta = h$, contrary to the irrationality of β.

A standard consequence of p-adic linear-form theory is that for fixed multiplicatively independent positive rationals α_1, α_2 and a fixed prime p, there are effectively computable constants C_0, C_1 such that

$$v_p(\alpha_1^d - \alpha_2^h) \le C_0 + C_1 \big(\log(2 + \max\{d, h\})\big)^2. \tag{91}$$

Much sharper versions, with linear dependence on $\log B$, are known in important two-logarithm regimes; the square is more than sufficient here [5, 6, 7]. Apply this to $(A, 3, p = 2)$ and $(M, 2, p = 3)$. Since $d, h = O_\beta(N)$, each factor in (83)–(87) has valuation $O_{M,A}((\log N)^2)$.

The number of pairs within the vertical fibers is

$$\sum_u \binom{s_u}{2} = \frac{L_N^3}{6\vartheta^2} + O_\vartheta(L_N^2) = O_\vartheta(N^{3/2}), \tag{92}$$

while the number within the horizontal fibers is

$$\sum_v \binom{\#U_v}{2} = \frac{L_N^3}{6\vartheta} + O_\vartheta(L_N^2) = O_\vartheta(N^{3/2}). \tag{93}$$

We obtain the following rigorous scale separation.

Theorem 8.2 (First tied layer is lower order)**.** *In the cross-coprime branch,*

$$v_2(I_{2,T}) + v_3(I_{3,T}) = O_{M,A}\Big(N^{3/2}(\log N)^2\Big) = O_{M,A}\Big(T^{3/2}(\log T)^2\Big). \tag{94}$$

In particular,

$$\frac{v_2(I_{2,T}) \log 2 + v_3(I_{3,T}) \log 3}{H_T} \longrightarrow 0. \tag{95}$$

Proof. Sum (91) over the pairs in (92) and (93), using (89)–(90). Finally use $N \asymp_\beta T$ and $H_T \asymp_\beta T^{5/2}$. $\square$

Status. [new paper proof] modulo the quoted classical p-adic linear-form estimate (91). The factorization of the tropical initial forms is elementary and is a particularly attractive Lean target.

8.4 What a leading excess would have to do

Theorem 8.2 does not prove $v_2(D_T) - \tau_{2,T} = o(H_T)$ or its 3-adic analogue. If the initial form is divisible by a power exceeding the first tropical gap, terms from the next valuation layer enter; those can cancel, exposing another layer, and so on. A leading-order excess is therefore possible only through a *cascade* of cancellations across a number of tropical layers growing with T. Cancellation confined to the minimum-permutation fibers is quantitatively negligible.

This yields a more precise version of the p-adic research target.

Research target 8.3 (No shallow-layer proof)**.** Either prove that the tropical cancellation cascade has total depth $o(H_T)$—which would retire the single-block p-adic strategy—or identify a structural identity forcing $\gg H_T$ cumulative cancellation across successively nonminimal assignments. Pairwise congruences such as (89) and (90) cannot by themselves supply the latter, because p-adic logarithm bounds make each congruence only polylogarithmic in the row separation.

9 Synthesis of the determinant audits

The preceding calculations put several superficially different determinant proposals into the same currency: the fraction of the leading archimedean budget supplied automatically by valuations of the entries. The numerical constants should not be compared as though the underlying determinants were identical—the ordinary, mixed-block, and complete-grid systems have different sizes and different upper bounds—but they do show which mechanism is closest to a contradiction before any genuine cancellation is used.

Three conclusions survive every normalization.

1. Merely adding every prime in the fixed support does not close any determinant. It does, however, improve the diagnosis substantially: the universal mixed-block ceiling is $4/5$, not a heuristic “about one half”.

2. The exact primitive theorem is quantitatively useful. The condition $\min(v_2(M), v_3(A)) = 0$ creates one structural slope vector of norm at least $4/15$; the conservation law forces a second norm contribution of at least the same size.

3. A successful p-adic proof must be a cancellation theorem, not a divisibility-by-every- term theorem. In the easiest branch, even the first large tied family contributes only a lower-order amount. The required phenomenon would have to repeat through many tropical layers or interact with a much sharper archimedean free energy.

Table 3: Leading termwise-divisor ceilings obtained in this continuation.

System	Ceiling or exact limit	Meaning
Ordinary input/output Vandermonde product	$1-\dfrac{\delta_2\log 2+\delta_3\log 3}{3\beta\log 6}<0.8710491$	Exact fixed-support divisor fraction; all primes already occurring in M and A included.
Mixed single block, arbitrary branch	$1-\dfrac{3}{8}\sum_p \mathfrak{G}(c_p)\le 4/5$	Exact all-prime tropical limit and universal primitive-depth ceiling.
Mixed single block, cross-coprime	$\le 7/10$	Uses $c_2=(-1,0)$, $c_3=(0,-1)$ and the external sum $(1,1)$.
Mixed single block, only structural primes in the cross-coprime branch	$3/10$	Exact contribution of $p=2,3$ before external primes or cancellation.
Complete triangular row grid	$\le \pi/4=0.7853982\ldots$	All-prime common divisor versus the sharp largest-Leibniz-term transport upper bound.
First tied fibers at 2 and 3	$o(1)$ of the mixed-block budget	Their valuations are $O(T^{3/2}(\log T)^2)$ against $H_T\asymp T^{5/2}$.

9.1 Correction of the post-attachment "one half / two thirds" note

After the attached unified report was produced, the repository acquired a short exploratory note, `Analysis/ExponentialIdentities/Article/new-reports/report-7.md`, at commit `ff9a9f5aa3a3f02599de93fc053e60918f269855`. That note correctly identified the distinction between termwise min-plus divisibility and cancellation beyond the tropical minimum, and its real/p-adic compatibility diagnosis remains valuable. It proposed, however, a structural $1/2$ ceiling and an all-prime $2/3$ ceiling. Neither constant is valid in the stated generality.

The structural averaging argument omitted the base and cross depths that appear in (51)–(52). Even under $v_2(M)=0$, the terms $v_2(A)kv$ and $v_3(M)ku+v_3(A)kv$ do not disappear. The exact structural contribution is obtained by retaining c_2 and c_3 in Theorem 6.3; it can exceed one half of H_T.

The all-prime step used the assertion that an oppositely sorted nonnegative triangular weight has assignment minimum at most two thirds of its mean. For the weight $Y=X+Z$ on the standard triangle, the exact values from Lemma 6.2 are

$$\mathbb{E}Y=\frac{2}{3},\qquad \Phi(1,1)=\frac{4}{15},\qquad \frac{\Phi(1,1)}{\mathbb{E}Y/2}=\frac{4}{5}.$$

Thus $4/5$, not $2/3$, is already attained by the relaxed external rank-one grading. The Gini-norm identity explains the general case and leads to the rigorous ceiling in Theorem 6.6. This is a correction of a calculation in an exploratory note, not a criticism of the source report itself; the attached unified report did not contain the claim.

Status. [new paper proof] — the correction follows from the exact transport formula, not from numerical experimentation.

10 Directed-rounding extension through 2^{27}

The attached report supplied a complete MPFR scan through 2^{26}. Because a finite extension is independent of the unresolved transcendence barrier and gives a useful stronger floor for every hypothetical generator, I extracted the literal C listing from its Appendix "Directed-rounding verifier", compiled it against MPFR 4.2.2, replayed an old audit chunk, and scanned the next full dyadic range.

10.1 Manifest audit

The literal extracted source has SHA-256 digest

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d`.

This does *not* equal the digest `ec709623f812478fbb7fb5567aa59fbfe3df0ae639ca366bc7c72e27 8bcfee2f` printed in the attached report. None of the ordinary line-ending, final-newline, or tab normalizations explains the discrepancy. The report therefore contains a stale or externally generated source-identity hash.

The semantic reproducibility check succeeds. Replaying the old interval $[29{,}360{,}128, 33{,}554{,}432)$ at 192 bits gives exactly the report's records

$$\text{lower: } 1.4439124872349507 \times 10^{-19} \text{ at } (30{,}103{,}832, 713{,}403{,}245{,}915),$$
$$\text{upper: } 5.4132120119800561 \times 10^{-21} \text{ at } (29{,}411{,}704, 687{,}581{,}893{,}443),$$

with zero failures. Thus the listing used here reproduces the old executable's mathematical behavior on an audited four-million-input overlap even though the printed file digest does not identify it.

Status. [computation] — this section is a directed-rounding software result. It is stronger numerically than the Lean theorem but is not a kernel proof. The manifest hash mismatch is explicitly part of the trust boundary.

10.2 The new range

The half-open interval $[2^{26}, 2^{27})$ was divided into sixteen chunks of width 2^{22}. The executable used 192-bit MPFR endpoints and outward rounding exactly as in the source report. There are 2^{26} integers in the range; the first, 2^{26}, is the sole power of two and is intentionally skipped. Hence the required nonpower count is $2^{26} - 1 = 67{,}108{,}863$.

The concatenated canonical extension log has SHA-256 digest

`b11843448cd8946442aba13758b5b666e4ac6eae50d2243fbdc32d83274cea8d`.

Every chunk reports zero failures, and the checked counts sum to the required total.

Table 4: Directed-rounding extension on $[2^{26}, 2^{27})$.

Chunk	Integer range	Checked	Smallest lower gap	Smallest upper gap
0	67,108,864–71,303,167	4,194,303	4.2837×10^{-20}	1.7723×10^{-19}
1	71,303,168–75,497,471	4,194,304	2.2961×10^{-20}	1.1433×10^{-20}
2	75,497,472–79,691,775	4,194,304	1.2813×10^{-19}	1.6063×10^{-20}
3	79,691,776–83,886,079	4,194,304	2.5420×10^{-20}	5.2012×10^{-20}
4	83,886,080–88,080,383	4,194,304	2.6166×10^{-20}	3.8496×10^{-20}
5	88,080,384–92,274,687	4,194,304	1.0923×10^{-19}	5.7144×10^{-20}
6	92,274,688–96,468,991	4,194,304	1.1017×10^{-20}	2.3227×10^{-20}
7	96,468,992–100,663,295	4,194,304	3.0759×10^{-20}	7.4810×10^{-20}
8	100,663,296–104,857,599	4,194,304	5.6465×10^{-20}	6.4744×10^{-20}
9	104,857,600–109,051,903	4,194,304	8.8144×10^{-20}	8.3662×10^{-21}
10	109,051,904–113,246,207	4,194,304	3.8892×10^{-20}	7.0327×10^{-21}
11	113,246,208–117,440,511	4,194,304	9.9365×10^{-21}	2.1814×10^{-20}
12	117,440,512–121,634,815	4,194,304	4.4964×10^{-20}	5.4132×10^{-21}
13	121,634,816–125,829,119	4,194,304	1.0774×10^{-20}	3.4093×10^{-20}
14	125,829,120–130,023,423	4,194,304	4.3609×10^{-20}	8.3073×10^{-22}
15	130,023,424–134,217,727	4,194,304	5.9082×10^{-21}	5.1326×10^{-20}
	total	**67,108,863**		

10.3 New extremal records and high-precision replay

The smallest lower logarithmic margin in the extension is

$$5.9082259992879571 \times 10^{-21} \quad \text{at} \quad (m, n) = (132{,}833{,}765, 7{,}501{,}348{,}219{,}569). \tag{96}$$

A 120-decimal-digit reevaluation gives

$$m^{\vartheta} - n = 6.39397545334704098376801255 10\ldots \times 10^{-8}.$$

The smallest upper logarithmic margin is the new global record

$$8.3073231820333117 \times 10^{-22} \quad \text{at} \quad (m, n) = (129{,}799{,}345, 7{,}231{,}571{,}292{,}851), \tag{97}$$

where

$$n + 1 - m^{\vartheta} = 8.66699043558189568532979522 88\ldots \times 10^{-9}.$$

Both one-input cases were rerun independently at 320 and 512 bits. The directed margins and nearest integer were unchanged in every displayed digit, and both replays reported zero failures.

Theorem 10.1 (Software-level finite exclusion through 2^{27})**.** *At the stated MPFR software-trust level, for every integer $1 \le m < 2^{27}$,*

$$m^{\vartheta} \in \mathbb{N} \quad \Longrightarrow \quad m \text{ is a power of two.}$$

Consequently every nonintegral real x satisfying $2^x, 3^x \in \mathbb{Z}$ obeys

$$x > 27. \tag{98}$$

Proof. The attached report's complete scan covers $1 \le m < 2^{26}$, and the sixteen new chunks cover $2^{26} \le m < 2^{27}$. In each case the directed logarithmic interval for $\vartheta \log m$ lies strictly between the intervals for the logarithms of two consecutive integers, except when m is a power of two. If $m = 2^x$ came from a nonintegral solution and $m < 2^{27}$, then $m^\vartheta = 3^x$ would be an integer at a non-power input, contradicting the scan. The endpoint $m = 2^{27}$ corresponds to the integral exponent 27, so a nonintegral solution must have $m > 2^{27}$ and hence $x > 27$. ☐

This raises the software-level spacing ratio in the optimized finite-difference comparison from $26 \log 3$ to $27 \log 3 = 29.662531\ldots$. It does not change any kernel-verified theorem; the formal floor remains $x \ge 12$ until the finite certificates are replayed in Lean.

11 Additional proof routes pushed to their failure points

The determinant calculations were the main new line of attack, but they were not the only one. This section records four other routes in enough detail to prevent the same hidden loss from being rediscovered under different notation.

11.1 Transferring convergents of β to ϑ

Let p/q be a continued-fraction convergent of a hypothetical primitive generator β and put

$$\varepsilon = q\beta - p.$$

For all sufficiently large q, $|\varepsilon| < 1/2$. Define the two nonzero integer gaps

$$D_2 = M^q - 2^p = 2^p(2^\varepsilon - 1), \qquad D_3 = A^q - 3^p = 3^p(3^\varepsilon - 1). \tag{99}$$

They manufacture an exact rational approximation to ϑ:

$$R_{p,q} := \frac{2^p D_3}{3^p D_2} = \frac{3^\varepsilon - 1}{2^\varepsilon - 1} \in \mathbb{Q}. \tag{100}$$

Taylor expansion at zero gives

$$R_{p,q} = \vartheta + \frac{\log 3\,(\log 3 - \log 2)}{2 \log 2}\,\varepsilon + O(\varepsilon^2), \tag{101}$$

so $|R_{p,q} - \vartheta| \ll |\varepsilon|$.

This looks promising because ϑ has an explicit irrationality measure. Wu's simultaneous linear-independence estimate for $1, \log 2, \log 3, \log 5$ gives, after setting the unused coefficients to zero, a conservative exponent

$$\left|\vartheta - \frac{r}{s}\right| \ge c_\vartheta s^{-\mu_\vartheta}, \qquad \mu_\vartheta = 8.616 \tag{102}$$

for all sufficiently large denominators s [8]. The obstacle is the denominator created by (100). Before reduction it is at most

$$3^p |D_2| \ll 3^p 2^p |\varepsilon| = 6^p |\varepsilon|. \tag{103}$$

Combining (101)–(103) with (102) gives only

$$|\varepsilon| \gg 6^{-p\mu_\vartheta/(1+\mu_\vartheta)}. \tag{104}$$

Since

$$6^{-\mu_\vartheta/(1+\mu_\vartheta)} = 2^{-2.316\ldots},$$

this is substantially weaker than the elementary bound $|\varepsilon| \gg 2^{-p}$ obtained directly from $D_2 \neq 0$. Sharper real irrationality measures cannot repair the architecture: every irrationality exponent is at least two, and the unreduced denominator already contains both 2^p and 3^p.

Proposition 11.1 (Exact denominator-loss diagnosis)**.** *The convergent-transfer construction can improve the elementary gap only if one proves an exponential lower bound for*

$$G_{p,q} = \gcd(2^p D_3, 3^p D_2) \tag{105}$$

that survives reduction of $R_{p,q}$. In the absence of such a bound, every known irrationality measure for ϑ is defeated by the factor 6^p in (103).

No mechanism forcing $G_{p,q}$ to be exponentially large was found. The two gaps share the same small real parameter ε, but real closeness does not impose a common prime divisor.

11.2 Two-row determinants and fixed-support S-units

For two block rows $i < j$, put $d = j - i$ and $h = n_i - n_j = \lfloor j\beta \rfloor - \lfloor i\beta \rfloor$. Their 2×2 determinant has the exact factorization

$$m_{T,i}a_{T,j} - m_{T,j}a_{T,i} = 6^{n_j}(MA)^i(2^h A^d - 3^h M^d). \tag{106}$$

Moreover

$$\frac{2^h A^d}{3^h M^d} = \left(\frac{3}{2}\right)^{\beta d - h}. \tag{107}$$

Thus a good rational approximation h/d to β makes the bracket in (106) a small *relative* difference of two integers supported on the fixed prime set dividing $6MA$.

Let

$$U = 2^h A^d, \qquad V = 3^h M^d, \qquad G = \gcd(U, V).$$

The reduced nonzero integer $(U - V)/G$ gives

$$|\beta d - h| \gg_{M,A} \frac{G}{\max\{U, V\}}. \tag{108}$$

The exponent of the right-hand side can be written explicitly as the positive part of a valuation-vector difference:

$$\log \frac{U}{G} = \sum_p \max\{h\mathbf{1}_{p=2} + dv_p(A) - h\mathbf{1}_{p=3} - dv_p(M), 0\} \log p. \tag{109}$$

As $h/d \to \beta$, this is $\kappa_{M,A}d + o(d)$ for a positive constant $\kappa_{M,A}$. Hence (108) is merely exponential in d, much weaker than the Baker–Matveev polynomial separation of Section 4.

The *abc* conjecture applied to the reduced equation $U/G - V/G = (U - V)/G$ would make the difference nearly as large as a fixed positive power of U/G, but it would still allow a relative gap $e^{-\epsilon d}$. Continued-fraction gaps of order $1/d$ are much larger. Thus this direct S-unit equation does not become a proof even after inserting *abc*; one would need an additional theorem controlling the prime support or radical of the difference.

11.3 Automaticity and the synchronized Sturmian word

The coding $s_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$ from Section 3 is aperiodic Sturmian. A standard theorem in automatic-sequence theory says that no aperiodic Sturmian word is automatic in any integer base [11, 12]. One quick obstruction is frequency: the frequency of 1 in s is the irrational number α, whereas an automatic sequence having a letter frequency has rational frequency.

This observation blocks a tempting but invalid shortcut. The rationality of $U = 2^{\alpha}$ does *not* make the threshold itinerary of $R \mapsto UR/2^{\mathbf{1}_{UR\geq 2}}$ finite-state; its exact rational denominators grow with time. The same is true for V in base three. Therefore Cobham's theorem cannot be invoked merely because the same word is produced by a dyadic and a triadic rational system. A viable automata proof would first have to manufacture genuine finite kernels in two multiplicatively independent bases, and doing so is already a new arithmetic theorem about U and V.

11.4 Real and p-adic logarithms

At the real place the counterexample is exactly

$$\log M \ \log 3 - \log A \ \log 2 = 0. \tag{110}$$

At $p = 2$ or 3, after stripping valuation and torsion parts, the units among $M, A, 2, 3$ admit p-adic logarithms. The determinant entries then have unit parts whose p-adic logarithms are bilinear forms in the row and column coordinates. This is the natural language for the cancellation cascade isolated in Section 8.

What is missing is a transfer principle from (110) to a congruence or rank drop among those p-adic logarithms. Real exponentiation by the transcendental number β does not define a compatible p-adic exponent, and the equality $M = 2^{\beta}$ has no p-adic meaning. Known p-adic linear-form theorems control a nonzero form once its integer coefficients and algebraic bases are specified; they do not create a p-adic form from a real one.

Research target 11.2 (Real/p-adic compatibility)**.** Find an algebraic construction, valid for the integer pair (M, A) alone, that converts (110) into either

1. a nontrivial congruence among p-adic unit logarithms with precision $\gg T$, repeated over $\gg T^{3/2}$ fiber pairs; or
2. a rank drop in the tropical initial matrices persisting through $\gg T$ valuation layers.

Without such a construction, ordinary p-adic logarithm bounds point in the opposite direction: they show that individual congruences are only polylogarithmically deep.

11.5 Transcendental-curve point counting

The graph $y = x^{\vartheta}$ is definable in the real exponential field, so Pila–Wilkie type results [16] give subpolynomial bounds for rational points outside algebraic pieces. Those generic bounds are far too weak here. The curve already contains the infinite sequence $(2^n, 3^n)$, giving $\asymp \log H$ integer points of height at most H, while a hypothetical second generator produces $\asymp (\log H)^2$ points globally. A proof by point counting would therefore need a sharp $O(\log H)$ theorem with a structural classification of the extremal sequence, not merely $O(H^{\epsilon})$. Such a theorem would essentially be another form of the rank-one conclusion sought by the conjecture.

12 Exact sufficient conditions extracted from the new analysis

A useful continuation should not stop at saying that the present determinant fails. The calculations above isolate numerical thresholds which, if crossed by a future theorem, would settle the conjecture. This section records those thresholds as precise implications.

12.1 The mixed-block cancellation-excess criterion

Let

$$\mathcal{P} = \{p : p \mid 6MA\}$$

be the fixed support of all determinant entries, and retain the mixed block determinant D_T, its tropical minima $\tau_{p,T}$, and the archimedean scale H_T. Define the support-prime cancellation excess

$$\mathcal{E}_T = \sum_{p\in\mathcal{P}} \left(v_p(D_T) - \tau_{p,T}\right) \log p \geq 0 \tag{111}$$

and the exact tropical deficit

$$\delta(M, A) = \frac{3}{8} \sum_{p\in\mathcal{P}} \mathfrak{G}(c_p). \tag{112}$$

Theorem 6.3 says $\log Q_T^{\mathrm{trop}} = (1 - \delta(M, A) + o(1))H_T$, while the elementary Leibniz upper bound is $\log D_T \leq H_T + o(H_T)$.

Theorem 12.1 (Cancellation-excess criterion)**.** *If a hypothetical primitive counterexample satisfies*

$$\liminf_{T\to\infty} \frac{\mathcal{E}_T}{H_T} > \delta(M, A), \tag{113}$$

then no such counterexample exists. In particular it is enough to prove an excess greater than $H_T/5$ uniformly in all primitive branches; in the cross-coprime branch the exact threshold is at least $3H_T/10$.

Proof. The selected support primes alone give

$$\log D_T \geq \sum_{p\in\mathcal{P}} v_p(D_T) \log p = \log Q_T^{\mathrm{trop}} + \mathcal{E}_T.$$

Under (113), the right-hand side exceeds H_T by a fixed positive multiple for all large T, contradicting the Leibniz upper bound. The numerical thresholds follow from Theorem 6.6 and Corollary 6.7. ☐

This criterion is not tautological: it involves only valuations at the fixed finite set of primes already present in $6MA$, whereas D_T can acquire arbitrary new prime factors. It states exactly how much same-support cancellation is required.

12.2 A denominator-cancellation criterion

For the transferred approximant $R_{p,q}$ of Section 11.1, let $s_{p,q}$ be its reduced denominator. If one could prove along infinitely many convergents that

$$s_{p,q} \ll q^{\eta} \qquad \text{for some } \eta < \frac{1 + \mu_\vartheta}{\mu_\vartheta}, \tag{114}$$

then Wu's measure would give $|R_{p,q}-\vartheta| \gg q^{-\eta\mu_\vartheta}$, whereas (101) and the convergent bound give $|R_{p,q}-\vartheta| \ll q^{-1}$. Since $\eta\mu_\vartheta < 1+\mu_\vartheta$ is not by itself enough, one tracks $|\varepsilon| < 1/q_+$ more precisely; a contradiction follows whenever $q_+ \gg q^{\eta\mu_\vartheta+\epsilon}$ along the same subsequence. A simpler sufficient condition is

$$s_{p,q} = q^{o(1)} \tag{115}$$

for infinitely many convergents with $q_+ \geq q^{1+\epsilon}$.

Equivalently, the gcd $G_{p,q}$ in (105) would have to cancel essentially the entire exponential factor $6^p|\varepsilon|$, leaving a subpolynomial denominator. This is a sharp formulation of the otherwise vague request for "correlation between the two integer gaps."

12.3 An automaticity criterion

The synchronized coding gives another exact, if presently inaccessible, implication.

Proposition 12.2 (Finite-kernel criterion)**.** *If under a hypothetical counterexample the common word $s_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$ were automatic in any integer base $q \geq 2$, then the Alaoglu–Erdős conjecture would follow.*

Proof. An automatic Sturmian word is ultimately periodic. Its slope, equal to the frequency of the letter 1, is therefore rational. But $\alpha = \beta - \lfloor \beta \rfloor$ would then be rational, and the rational-exponent theorem would make β integral. □

The missing step is not symbolic dynamics; it is proving automaticity from the two rational multipliers U and V. The growing denominators in (5) show exactly why the obvious state space is infinite.

13 Branch-sensitive research program

The source report already classifies a hypothetical pair much more sharply than "two integers on a curve." The next attack should exploit that classification rather than seek a single estimate that treats every branch identically.

13.1 Branch I: independent external valuation directions

Here the second-iterate kernel K_β is zero, equivalently the external prime-valuation vectors of M and A are independent. Corollary 6.5 gives a strict angular defect $\Delta_{\text{ext}} > 0$ in the all-prime tropical fraction. That makes the termwise divisor farther from the archimedean upper bound, but the same independence also supplies at least two genuinely different external primes at which cancellation can be studied.

The concrete target is a simultaneous p-adic rank theorem for the tropical initial matrices: show that if cancellation persists through L layers at two non-collinear external primes, then an integer relation appears among the two external valuation vectors. Such a theorem would contradict $K_\beta = 0$. Existing one-prime linear-form bounds do not compare the residue kernels at different primes; this is the genuinely new ingredient required.

13.2 Branch II: a common external valuation ray

When $K_\beta \neq 0$, all external vectors are collinear and the source report gives the exact Möbius-orbit condition

$$\beta = \frac{u + v\vartheta}{r + s\vartheta} \qquad (u, v, r, s \in \mathbb{Q}, us - vr \neq 0). \tag{116}$$

This is the branch in which the external triangle inequality can be an equality. It is also arithmetically lower-dimensional: after clearing denominators, the external parts of M and A are powers of one common primitive core.

The best next step is therefore not a generic all-prime determinant. It is a mixed real/radical invariant built from the canonical core and the two binomial fields already formalized in the repository. Useful intermediate targets are:

1. compare the field norms of the canonical radicals on the two sides of (116);
2. show that simultaneous radical degrees force an additional common perfect-power degree, contradicting affine primitivity; or
3. prove that the Möbius coefficients in (116) cannot have the sign pattern imposed by nonnegative prime valuations unless one output is $\{2, 3\}$-smooth.

The latest repository modules `CanonicalRadicalFieldNorm`, `CanonicalRadicalValuation`, `CanonicalRadicalSharpness`, `CanonicalRadicalDivisibleHull`, and `SimultaneousRadicalDegrees` provide much of the algebraic plumbing for this branch.

13.3 Branch III: one smooth output

If M is $\{2, 3\}$-smooth, then $\beta = a + d\vartheta$ with $d > 0$; if A is smooth, there is a symmetric affine formula. The source corpus proves that both outputs cannot be smooth and that the other output contains a prime at least five. In the extreme case $M = 3^d$, the problem is the localized radical assertion

$$(3^d)^\vartheta \notin \mathbb{Z}[1/3].$$

This branch is the smallest transcendence-theoretic wall and deserves separate treatment. Potentially useful objects are the irreducible binomial for the least localized radical, its field norm, and reduction at primes over two and three. A successful local argument must remain sensitive to the real identity defining ϑ; ordinary ideal factorization alone is already exhausted by the canonical-radical modules.

13.4 Branch IV: cross-coprime primitive outputs

The cross-coprime branch has the strongest determinant constants and the cleanest residue matrices. Here the exact first tropical forms are the fiber Vandermondes of Section 8. This is the best laboratory for determining whether the cancellation cascade is real or illusory.

The next computational experiment should not search for counterexamples. For synthetic odd pairs (M, A) and Beatty depths generated by $\beta = \log_2 M$, compute

$$v_2(D_T) - \tau_{2,T}, \qquad v_3(D_T) - \tau_{3,T}$$

for increasing T, together with the histogram of tropical layers contributing at the final valuation. Small exact experiments performed during this work show an excess compatible with $N^{3/2}$ rather than $N^{5/2}$, matching Theorem 8.2; they are reconnaissance only, since a synthetic A does not satisfy $A = 3^\beta$. A proof should seek a uniform $o(H_T)$ bound, probably through p-adic exponential-polynomial zero estimates.

13.5 Archimedean priority: solve the triangular HCIZ limit

The complete grid finally supplies the missing T^2 rows, and its empirical measures are explicit. This makes Target 7.6 the highest-priority analytic problem. The required calculation is a large-N asymptotic for the HCIZ integral with both spectra having density $2t$ on $[0,1]$, under the coupled scaling $x_iy_j \asymp T^2$ and $N \asymp T^2$. Even a nonsharp upper bound improving the largest-term energy by more than $1-\pi/4$ would materially change the p-adic comparison.

A practical sequence is:

1. numerically evaluate $\log\det(e^{x_iy_j})$ using high-precision total-positive algorithms for moderate T;
2. subtract the exact Vandermonde/factorial terms in (80) to estimate the free-energy limit;
3. derive the corresponding variational problem using known HCIZ large-deviation methods;
4. only then reinsert the primewise valuation lower bounds.

This route is attractive because the analytic limit does not depend on the unknown prime factorization of M and A; only β enters the overall scaling.

13.6 Finite verification priority

The software floor is now $x > 27$. The immediate formal target is not another raw doubling of the MPFR scan but a replayable certificate architecture. A realistic hierarchy is:

1. a fast untrusted generator records only near-boundary pairs and rational interval data;
2. a small verified checker proves each interval enclosure and the monotone coverage between records;
3. chunks are combined by a theorem whose statement mentions only integer endpoints;
4. the final theorem is imported into the aggregate surface without `native_decide` or an opaque external axiom.

The source-hash discrepancy found in Section 10 makes this especially worthwhile: mathematical replay is more robust than a manifest that identifies an external binary only by prose.

14 Lean formalization blueprint

The source corpus is already unusually strong on exact conditional structure. The most useful continuation is therefore not a second informal reimplementation of those results, but a small collection of sharply separated modules whose statements expose the genuinely new invariants. This section gives a dependency order designed to keep arithmetic, asymptotics, and transcendence inputs from leaking into one another.

14.1 Proposed module graph

Proposed module	Principal statement	Main dependencies
`SturmianNormalization`	Exact recurrences for R_k, S_k, common jump word s_k, and the exponent/output correspondence.	Existing primitive-generator and denominator-normalization modules; elementary floor arithmetic.
`ReducedOrbitDenominators`	Lowest-term denominators $q_k^{(2)}, q_k^{(3)}$ and the full-depth alternative from $\min(v_2(M), v_3(A)) = 0$.	`SturmianNormalization`; unique factorization and valuations.
`EffectiveGeneratorSeparation`	An abstract interface turning a two-logarithm lower bound into $\lvert\beta - p/q\rvert \geq cq^{-\mu}$ and a polynomial recurrence for continued-fraction denominators.	A stated external Baker–Matveev hypothesis, plus existing continued-fraction lemmas.
`BlockVandermondeValuation`	Exact v_2 and fixed-support v_p formulas for the conditional block Vandermonde; output analogue.	Existing exact block parametrization; valuation of a difference with unequal valuations.
`OrdinaryAdelicFraction`	The asymptotic fraction (32) and the universal $1 - \log 2/(3 \log 6)$ ceiling.	`BlockVandermondeValuation`; polynomial floor sums; primitive-output depth theorem.
`TriangleGini`	$\mathfrak{G}$ is a norm and satisfies the explicit formula (48).	Elementary measure integration, or a finite polygonal integration library.
`FiniteTropicalAssignment`	Exact rearrangement formula for a finite row list and a finite column-weight list.	Sorting/permutations; finite rearrangement inequality.
`MixedColumnRiemannSums`	Lattice count (34) and convergence of scaled low-weight columns to uniform measure on $\mathcal{T}$.	Two-dimensional lattice-point estimates and Riemann sums.
`MixedAdelicGini`	The limit (55), conservation $\sum_p c_p = 0$, and branch-sensitive three-vector bound.	Previous three modules; finite prime support.
`MixedTropicalCeilings`	Universal $4/5$, cross-coprime $7/10$, and structural $3/10$ constants.	`MixedAdelicGini`; primitive-output depth theorem; explicit values from `TriangleGini`.
`FullGridTransport`	Row/column empirical distributions, countermonotone energy $\pi/8$, maximum energy $1/2$, and the ratio $\pi/4$.	Triangular lattice sums, quantile integration, rearrangement.

Proposed module	Principal statement	Main dependencies
`TropicalInitialFibers`	Factorization of the first 2- and 3-adic initial forms into ordinary Vandermondes on column fibers.	Determinants over valued rings; stable sorting by valuations.
`InitialFiberValuationBound`	Under a standard p-adic logarithm hypothesis, the first tied layer contributes $o(H_T)$.	`TropicalInitialFibers`; an abstract two-logarithm valuation bound.
`FiniteCheck2Pow27`	A kernel-replayable certificate excluding non-powers of two below 2^{27}.	Existing finite-check certificate architecture; generated chunk certificates.

The first two modules are almost entirely combinatorial. The determinant modules should not import transcendence theory. Conversely, the Baker and p-adic logarithm interfaces should be abstract propositions whose classical realization is documented outside the kernel unless and until Mathlib contains a suitable theorem. This makes the trust boundary explicit.

14.2 Finite statements before asymptotics

The continuum notation in Sections 6 and 7 is compact, but Lean should first see finite inequalities. For rows $0 \le k < N$ and columns $z_j \in \mathbb{R}$, the finite rearrangement lemma should assert

$$\min_{\sigma \in S_N} \sum_{k=0}^{N-1} k z_{\sigma(k)} = \sum_{k=0}^{N-1} k z^{\uparrow}_{N-1-k}, \tag{117}$$

where $z^{\uparrow}$ is the nondecreasing rearrangement. Every tropical assignment in this report is an affine rescaling of (117). The limiting Gini identity can then be obtained by three auditable steps:

1. prove convergence of empirical distributions for the triangular lattice;
2. prove convergence of the first moments and pairwise absolute-difference moments;
3. invoke the finite identity

$$\frac{1}{N^2} \sum_{i,j} |z_i - z_j| = \frac{2}{N^2} \sum_{j=0}^{N-1} (2j - N + 1) z^{\uparrow}_j.$$

This route avoids formalizing optimal transport as a general theory.

14.3 Exact theorem sketches

The intended public statements can be phrased close to the mathematics.

```
/-- A primitive counterexample determines one common lower mechanical word
    driving both normalized rational orbits. -/
theorem exists_synchronized_sturmian_data
    (hfail : ExistsNonintegralSolution) :
    Exists fun B : Nat => Exists fun alpha U V : Real =>
      0 < alpha /\ alpha < 1 /\ Irrational alpha /\
```

```
      U = 2 ^ alpha /\ V = 3 ^ alpha /\
      (forall k : Nat,
        let e := Int.floor (k * alpha)
        normalizedTwo k = U ^ k / 2 ^ e /\
        normalizedThree k = V ^ k / 3 ^ e) := by
  ...

/-- The automatic all-prime divisor of the balanced mixed determinant
    cannot fill more than four fifths of its leading height. -/
theorem mixed_tropical_divisor_limsup_le_four_fifths
    (hprimitive : min (vTwo M) (vThree A) = 0) :
    limsup (fun T => log (tropicalDivisor T) / leadingHeight T)
      <= (4 : Real) / 5 := by
  ...

/-- The complete conditional grid has countermonotone/maximal transport
    ratio pi/4. -/
theorem full_grid_transport_ratio :
    countermonotoneEnergy / maximalEnergy = Real.pi / 4 := by
  ...
```

The actual APIs will need natural-valued denominator exponents rather than casts of negative integers, and the asymptotic statements will likely use `Tendsto` and `IsLittleO` rather than a custom `limsup`. The sketches are included to fix the mathematical interfaces, not Lean syntax.

14.4 Formalizing the first tied layer

A useful reusable notion is the initial form of a determinant at a prime. For a matrix over $\mathbb{Z}$ and an assignment minimum τ_p, reduce

$$p^{-\tau_p} \det(E_{ij}) \pmod{p}.$$

Only minimizing permutations survive. In the cross-coprime branch the minimizing permutations preserve equal-u fibers at $p = 2$ and equal-v fibers at $p = 3$; the surviving sum is a product of smaller determinants. This should be proved as a purely finite combinatorial determinant identity. The later valuation estimate can then be stated separately:

$$v_2(I_{2,T}) + v_3(I_{3,T}) = O\Big(N^{3/2}(\log N)^2\Big). \tag{118}$$

The big-O exponent is not sacred; any $o(T^{5/2})$ theorem is sufficient for the qualitative conclusion.

14.5 Certificate architecture for the new finite bound

The existing verified bound through 4096 should remain the trusted baseline until a new certificate is accepted. A practical route to 2^{27} is hierarchical:

1. partition $[1, 2^{27})$ into fixed-width chunks;
2. for each non-power-of-two m, store an integer nearest-value locator n_m and rational lower/upper enclosures proving either $m^\vartheta < n_m$ or $m^\vartheta > n_m$;
3. hash each chunk and prove a small checker correct in Lean;
4. combine the chunk theorems into one interval theorem.

The MPFR run in Section 10 is reconnaissance and a source of extremal cases; it is not the certificate itself.

15 Numerical sanity checks and reproducibility

None of the limiting constants in the determinant theorems depends on numerical evidence. The computations below were used to catch normalization errors, check convergence rates, and stress-test the closed forms before writing the proofs.

15.1 Balanced-block assignment constants

For each N, take the first N pairs (u, v) ordered by $u + \vartheta v$, pair them against N equally spaced row parameters, and solve the one-dimensional assignment problems by sorting. In the cross-coprime model, the structural places should tend to 3/10 and the minimal all-prime three-vector model to 7/10.

N	structural $2, 3$ fraction	all-prime three-vector fraction
25	0.3224571184	0.7258906037
50	0.3135103379	0.7141968856
100	0.3050363664	0.7053583986
250	0.3024422068	0.7025311250
500	0.3012614628	0.7013136826
1000	0.3006780555	0.7006854844
2500	0.3002789586	0.7002894348
5000	0.3001354539	0.7001384826
Limit	3/10	7/10

Table 6: Finite assignment sums converging to the exact mixed-block constants.

The convergence from above is consistent with the $O(L^{-1}) = O(N^{-1/2})$ boundary error in triangular lattice counting. The table also exposes the error in the proposed 2/3 ceiling: the finite all-prime ratio is already near 0.70, and the exact limit is 7/10.

15.2 Complete-grid transport ratio

For a representative hypothetical height parameter $\beta = 27.314159$, form the row lattice $n + \beta k \leq T$ and the matching low-weight column lattice. Normalize the countermonotone assignment by the largest-term assignment. The finite ratios are:

The slow boundary convergence is expected: both empirical triangles have $O(T)$ boundary points against $O(T^2)$ bulk points, and the matching weight scale also grows with T.

15.3 Directed-rounding scan manifest

The scan in Section 10 used the literal C listing extracted from the attached LaTeX source. Its SHA-256 digest is

T	number of rows	transport ratio
50	73	0.7415113
100	237	0.7559915
200	841	0.7671191
400	3157	0.7750371
800	12210	0.7796781
Limit	—	$\pi/4 = 0.7853981634\ldots$

Table 7: Finite full-grid countermonotone/maximal transport ratios.

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d`.

This differs from the digest printed in the old report. To ensure the discrepancy was only a manifest error, the extracted source was first rerun on the old overlap interval $[29{,}360{,}128, 33{,}554{,}432)$; it reproduced the old extremal records and reported zero failures.

The extension interval $[2^{26}, 2^{27})$ was split into sixteen chunks of width 2^{22} and run at 192 bits. The concatenated chunk log has SHA-256

`b11843448cd8946442aba13758b5b666e4ac6eae50d2243fbdc32d83274cea8d`.

The two new extremal records were replayed at 320 and 512 bits. A representative build and invocation sequence was:

```
cc -O3 -std=c17 mpfr_scan.c \
  /lib/x86_64-linux-gnu/libmpfr.so.6 -lgmp -lm -o mpfr_scan

./mpfr_scan 67108864 71303168 192
# repeat consecutive width-2^22 chunks through 134217728
sha256sum extension-through-2pow27.log
```

The scanner excludes powers of two from the tested population because they are the known solutions. Across the extension it checked 67,108,863 non-powers of two and reported zero failures.

15.4 Extremal replay records

The narrowest new lower-directed separation was

$$m = 132{,}833{,}765, \qquad n = 7{,}501{,}348{,}219{,}569, \qquad \text{directed margin } 5.9082259992879571 \times 10^{-21}.$$

The corresponding high-precision ordinary difference is approximately

$$m^{\vartheta} - n = 6.39397545334704 \times 10^{-8}.$$

The narrowest new upper-directed separation was

$$m = 129{,}799{,}345, \qquad n = 7{,}231{,}571{,}292{,}851, \qquad \text{directed margin } 8.3073231820333117 \times 10^{-22},$$

with

$$n + 1 - m^{\vartheta} = 8.666990435581896 \times 10^{-9}.$$

These decimal differences are explanatory only. The certification decision uses the MPFR outward-rounded interval endpoints and the scanner's locator checks.

16 Status matrix and dependency audit

Statement	Status	Dependency or caveat
Exact primitive monoid $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ and candidate monoid	[Lean]	Inherited from the attached report.
Primitive-depth condition $\min(v_2(M), v_3(A)) = 0$	[Lean]	Inherited; central to both universal ceilings.
Synchronized dyadic–triadic Sturmian recurrence	[new paper proof]	Elementary from the exact monoid and floor normalization.
Lowest-term denominator formulas and full-depth alternative	[new paper proof]	Unique factorization plus primitive depth.
Effective polynomial irrationality measure for fixed M	[classical] plus [new paper proof]	Baker–W"ustholz/Matveev; constants depend effectively on M.
Polynomial recurrence of continued-fraction denominators	[new paper proof]	Standard convergent inequality applied to the previous row.
Exact ordinary block valuations at all input-support primes	[new paper proof]	Unequal valuation of the two monomials in every difference.
Ordinary product fraction (32)	[new paper proof]	Exact floor sums; output endpoint errors are $O(T^2)$.
Universal ordinary ceiling $0.871049\ldots$	[new paper proof]	Primitive depth only; not a bound on cancellation from new primes in differences.
Explicit triangular Gini norm	[new paper proof]	Direct integration.
Mixed all-prime tropical limit (55)	[new paper proof]	Rearrangement plus triangular Riemann sums.
Universal $4/5$ mixed ceiling	[new paper proof]	Conservation, Gini triangle inequality, primitive depth.
Cross-coprime $7/10$ and structural $3/10$	[new paper proof]	Specialization $a = b = c = d = 0$.
Full-grid $\pi/4$ ceiling	[new paper proof]	Countermonotone and comonotone transport for the common quantile $\sqrt{t}$.
HCIZ determinant identity	[classical]	Exact finite identity; the needed large-N inequality remains open.
First tied-layer fiber factorization	[new paper proof]	Exact initial-form determinant algebra in the cross-coprime branch.

Statement	Status	Dependency or caveat
First tied layer is $o(H_T)$	[classical] plus [new paper proof]	Standard p-adic two-logarithm estimates; no claim about higher valuation layers.
Mixed cancellation-excess criterion	[new paper proof]	Immediate sufficient condition from integrality and the Gini deficit.
Software exclusion through 2^{27}	[computation]	Complete MPFR directed-rounding scan; not a Lean proof.
Kernel exclusion through 4096	[Lean]	Remains the strongest formal finite theorem audited in the source report.
The Alaoglu–Erdős conjecture itself	[open]	Equivalent to the unresolved special four-exponentials determinant.

16.1 Logical dependency graph

The new deductions can be summarized as

$$\begin{array}{c} \text{exact primitive monoid} \ + \ \text{primitive depth} \\ \Downarrow \\ \text{synchronized Sturmian system} \qquad \text{and} \qquad \text{exact block valuation profiles} \\ \Downarrow \\ \text{ordinary all-prime fraction} \\ \text{mixed Gini conservation law} \\ \text{full-grid transport problem} \\ \Downarrow \\ 0.871049\ldots \ \text{ordinary ceiling}, \quad 4/5 \ \text{mixed ceiling}, \quad \pi/4 \ \text{full-grid ceiling} \\ \Downarrow \\ \text{only cancellation beyond tropical minima or a sharper HCIZ bound can finish these routes.} \end{array}$$

Baker–Matveev separation and the finite scan are parallel inputs: they sharpen the possible counterexample but do not enter the determinant ceiling proofs.

17 Conclusions

The Alaoglu–Erdős conjecture remains unproved. The continuation nevertheless changes the shape of the search in several concrete ways.

First, the exact conditional monoid has a compact symbolic normal form. After removing the integer part of the primitive generator, one irrational Sturmian word controls both the binary and ternary denominator jumps of two rational recurrences. This exposes a possible route through finite-state rigidity, but it also prevents a false shortcut: rational multipliers do not by themselves make the word automatic.

Second, ordinary transcendence theory gives more Diophantine separation than the source report's elementary argument records. A fixed hypothetical generator is a ratio of two logarithms of integers, so Baker–W"ustholz/Matveev yields an effective polynomial irrationality measure. Continued-fraction denominators therefore grow at most polynomially from one stage to the next. This is useful quantitative structure, but it does not conflict with the $O(T)$ exact population of a dyadic block.

Third, the most direct p-adic determinant plan can now be audited with exact constants. For ordinary block Vandermondes, every valuation at every prime already supporting the inputs is accounted for, and primitive depth leaves at least $12.8950\ldots\%$ of the leading archimedean budget uncovered. For the balanced mixed determinant, all primes collapse into a finite family of slope vectors summing to zero. The triangular Gini norm then proves that entrywise divisibility misses at least 20% universally, at least 30% in the cross-coprime branch, and at least 70% if one uses only the structural places $2, 3$ there. The complete two-dimensional grid improves the geometry but still leaves a termwise ceiling of $\pi/4$.

These are no-go theorems for particular mechanisms, not evidence that the conjecture is false. They identify the exact remaining mechanisms:

1. a leading-order cascade of cancellations among many equal-valuation Leibniz terms;
2. an archimedean determinant estimate substantially smaller than its largest-term bound, most naturally through the HCIZ free energy;
3. a genuinely mixed real/p-adic construction that turns $\log M \log 3 = \log A \log 2$ into deep congruences; or
4. a rigidity theorem for the synchronized rational Sturmian system.

The first cancellation layer is provably too small, so any successful p-adic proof must propagate through many valuation layers rather than extracting one convenient Vandermonde factor.

Finally, the directed-rounding computation was extended by one full binary exponent. At the software-trust level, no non-power-of-two candidate occurs below 2^{27}, so a counterexample would satisfy $x > 27$. The formal Lean boundary remains $x \geq 12$ until a generated certificate is checked by the kernel. The old scanner-source hash printed in the attached report is incorrect; the literal source, overlap replay, new log hash, and high-precision extremal replays are recorded explicitly in this document.

Bottom line. The conjecture is still open, but the continuation produces new exact mathematics rather than only a list of ideas: a synchronized Sturmian equivalence; effective polynomial separation; closed all-prime Vandermonde asymptotics; a triangular Gini norm governing mixed tropical divisibility; universal 4/5, 7/10, 3/10, and $\pi/4$ ceilings; a first-layer cancellation theorem; exact sufficient thresholds for future work; and a reproducible software exclusion through 2^{27}. The next decisive calculation is the triangular HCIZ free energy or, on the arithmetic side, a proof of a genuinely leading tropical cancellation cascade.

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